UNIT 1

Introduction to Quantum Theory and Technologies

Unit 1 at a glance

1. Classical → quantum transition 2. Superposition, entanglement, uncertainty, wave-particle duality

3. Classical vs quantum mechanics 4. States & measurement 5. Electrons, photons, atoms

6. Quantization 7. Why quantum? 8. Quantum computing/communication/sensing 9. National quantum missions

1. From Classical to Quantum Physics

Classical physics (Newton, Maxwell, Boltzmann) explains the everyday world: falling apples, planets, heat engines, light as waves. By the late 1800s many physicists felt physics was 'almost complete'. But a few small puzzles refused to fit - and they cracked the whole picture open.

» The puzzles classical physics could not solve

- *Black-body radiation: a hot object's colour/spectrum. Classical theory predicted infinite energy at short wavelengths (the 'ultraviolet catastrophe'). Wrong!*
- *Photoelectric effect: light knocks electrons out of metal only if its frequency is high enough - however bright the light is. Waves could not explain this.*
- *Atomic stability & spectra: why do atoms emit light only at sharp, fixed colours (lines)? Why don't electrons spiral into the nucleus?*
- *Specific heat of solids at low temperature also disagreed with classical predictions.*

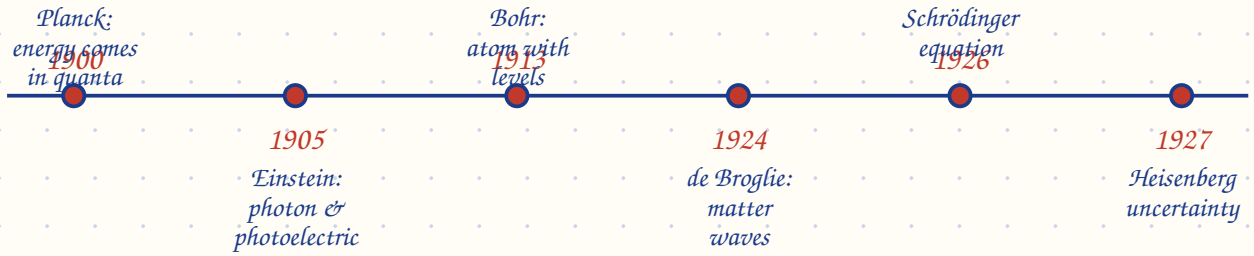


Fig 1: Key milestones in the birth of quantum theory

» The big ideas that fixed it

- *Max Planck (1900): energy is emitted/absorbed in tiny packets called quanta, $E = h\nu$, where $h = 6.626 \times 10^{-34}$ J·s is Planck's constant.*
- *Albert Einstein (1905): light itself is made of packets - photons, each with energy $E = h\nu$. Explained the photoelectric effect (Nobel Prize 1921).*

- *Niels Bohr (1913)*: electrons in atoms occupy only certain allowed energy levels; they jump between them by emitting/absorbing photons.
- *Louis de Broglie (1924)*: every particle has a wavelength $\lambda = h/p$ (matter waves).
- *Schrödinger, Heisenberg, Born, Dirac (1925-28)*: a complete mathematical theory - quantum mechanics.

> ★ *Remember*
> Quantum mechanics is the theory of the very small (atoms, electrons, photons) where energy comes in discrete packets and nature is described by probabilities, not certainties.

2. Fundamental Principles (Conceptual)

» 2.1 Wave-Particle Duality

Light and matter behave sometimes like *waves* (spreading, interfering) and sometimes like *particles* (arriving in lumps at a point). Which behaviour you see depends on how you set up the experiment. This is not a contradiction - quantum objects are simply a new kind of thing that is neither a classical wave nor a classical particle.

- *Wave evidence*: interference and diffraction (Young's double-slit with light; also seen with electrons, atoms, even large molecules).
- *Particle evidence*: photoelectric effect, Compton scattering, single dots on a detector screen.
- *Double-slit experiment*: send electrons one at a time through two slits. Each lands as one dot (particle), yet after thousands of dots an interference pattern (wave) appears. If you detect which slit each electron used, the pattern vanishes!



Fig 2: Double-slit experiment - waves interfere, detection makes dots

> *de Broglie wavelength*
> $\lambda = h / p = h / (m\,v)$
> Bigger momentum → smaller wavelength. A cricket ball's λ is ~10^{-34} m (unobservable) but an electron's λ is comparable to atom size - so electrons show wave effects. This is why quantum effects are 'hidden' in daily life.

» 2.2 Superposition

A quantum system can exist in a combination of several states at once. Example: an electron's spin can be 'up' AND 'down' simultaneously, written $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$ where $|\alpha|^2$ and $|\beta|^2$ are the probabilities of finding up or down when measured, and $|\alpha|^2 + |\beta|^2 = 1$.

- It is NOT that we are ignorant of which state it is - the system genuinely has no single definite value yet.
- Superposition is what makes interference possible and is the raw resource for quantum computing (qubits).
- Schrödinger's cat (thought experiment): a cat linked to a radioactive atom is 'alive + dead' until observed. Used to point out how strange superposition is when scaled up.

» 2.3 Entanglement

Two or more particles can be linked so that their properties are correlated no matter how far apart they are. Measuring one instantly tells you about the other. Einstein called it 'spooky action at a distance'. Example: two photons prepared so that their polarizations are always opposite; each alone is random, but together they are perfectly correlated.

- The whole system has a definite state, but the individual parts do not.
- Entanglement cannot be used to send messages faster than light (the individual results are random) - no-signalling.
- Bell's theorem + experiments (Aspect, Clauser, Zeilinger - Nobel 2022) showed nature really is non-local; no 'hidden variable' classical explanation works.
- Entanglement = key resource for quantum teleportation, quantum cryptography and quantum computing speed-ups.

» 2.4 Heisenberg's Uncertainty Principle

Statement

$\Delta x \cdot \Delta p \geq \hbar/2$ ($\hbar = h/2\pi$)

Also: $\Delta E \cdot \Delta t \geq \hbar/2$

You cannot know both the position and momentum of a particle with perfect precision at the same time. Squeezing one makes the other fuzzier.

- It is a basic property of nature (wave-like behaviour), NOT merely a flaw of our instruments.
- Position-momentum, energy-time, and spin components along different axes are such 'incompatible' pairs.
- Consequences: atoms are stable (electron can't collapse into the nucleus), zero-point energy exists, and quantum cryptography is secure (measuring disturbs the state).

Exam tip / Remember

The four pillars to learn by heart: Superposition, Entanglement, Uncertainty, Duality. Give one definition + one example for each - this is a common 5-mark question.

3. Classical vs Quantum Mechanics - Theoretical Comparison

Feature	Classical mechanics	Quantum mechanics
Scale	Everyday, large objects; planets	Atoms, electrons, photons, small scales
State described by	Position & momentum (definite values)	Wavefunction / state vector $\|\psi\rangle$
Predictions	Deterministic: same start → same result	Probabilistic: only probabilities of outcomes
Energy	Continuous - any value	Often discrete (quantized)
Measurement	Does not disturb the system (ideal)	Measurement changes the state (collapse)
Nature of matter	Particles OR waves	Wave-particle duality
Combination of states	Object is in one state at a time	Superposition of many states
Correlations	Local; due to common cause	Entanglement; non-local correlations
Equation	Newton: $F = ma$	Schrödinger: $i\hbar\, \partial\psi/\partial t = \hat{H}\psi$
Limit	—	Reduces to classical when action $\gg \hbar$ (correspondence principle)

Correspondence Principle (Bohr)

For large systems/quantum numbers, quantum predictions merge smoothly into classical ones. So quantum theory does not 'throw away' Newton - it contains it as a special limit.

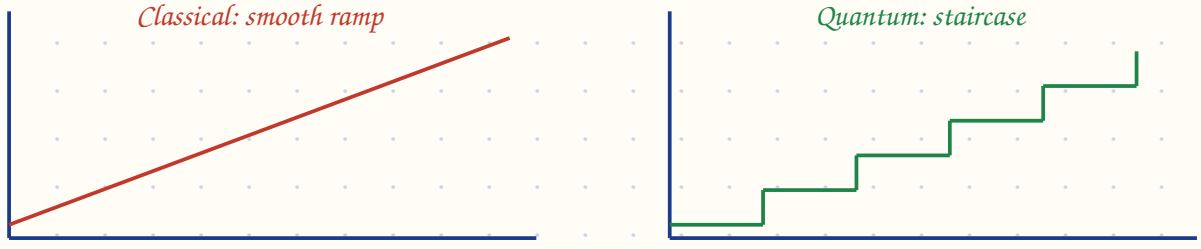


Fig 3: Classical = continuous ramp; quantum = discrete staircase

4. Quantum States and Measurement

» 4.1 What is a quantum state?

A quantum state is the complete description of a quantum system, written as a vector $|\psi\rangle$ (Dirac 'ket' notation). It carries all the information that can be known. For a two-level system: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are called probability amplitudes (complex numbers).

- Born rule: probability of outcome = $|\text{amplitude}|^2$.
- Amplitudes can add or cancel (interference) - probabilities from classical chance cannot cancel, which is the big difference.
- The state evolves smoothly and deterministically between measurements (Schrödinger equation).

» 4.2 The nature of observation (measurement)

- Before measurement: system may be in a superposition.
- During measurement: we interact with the system using a device and get ONE definite result (e.g. 'up').
- After measurement: the state is now the one we saw - this sudden change is called collapse of the wavefunction.
- Which result we get is random; only the probabilities are predictable. Repeating on identical systems gives a statistics.

Measurement - key points

- Measuring is an active process, not passive 'looking'.
- Measuring one property can destroy information about an incompatible one (uncertainty principle).
- 'Observer' does NOT need to be a human - any interaction that records the information acts as a measurement.
- Different interpretations exist (Copenhagen, Many-Worlds, Pilot-wave, Decoherence) - they differ in explanation, not in predictions.

- No-cloning theorem: an unknown quantum state cannot be copied perfectly. This protects quantum communication from eavesdropping.
- Decoherence: interaction with the environment effectively 'measures' the system, turning superposition into ordinary mixtures - why large objects look classical.

5. Overview of Quantum Systems

System	Quantum properties used	Where it is used
Electron	Spin (↑/↓), charge, energy levels; shows interference	Quantum dots, spin qubits, electron microscopes, semiconductors, transistors
Photon	Polarization (H/V), path, time-bin; travels fast, weakly interacts	Quantum communication, QKD, photonic computing, quantum optics
Atom / Ion	Discrete electronic energy levels, nuclear spin	Atomic clocks, trapped-ion quantum computers, laser cooling, sensors

» Electrons

Electrons are charged particles with an intrinsic property called spin (spin-½): it has two measurable values, 'up' and 'down' along any chosen axis. Electrons in atoms and solids occupy discrete energy states and obey the Pauli exclusion principle (no two electrons in the same state), which gives the periodic table and the structure of matter.

» Photons

A photon is a quantum of light: massless, travels at c, energy $E = h\nu = hc/\lambda$, momentum $p = h/\lambda$. Its polarization (direction of the electric field) gives a natural two-level system (horizontal/vertical). Photons barely interact with the environment, so they carry quantum information over long distances through optical fibre or space.

» Atoms

Atoms have electrons in discrete orbitals; transitions between levels absorb/emit photons of exact frequency, giving spectral lines. This exactness is exploited in atomic clocks (the SI second is defined from caesium-133 transitions at 9,192,631,770 Hz). Cooled, trapped atoms/ions are among the best qubits and sensors.

6. The Concept of Quantization

Quantization means that certain physical quantities can take only specific discrete values instead of any value in a continuous range. Energy of bound systems (atom, molecule, particle in a box, harmonic oscillator), angular momentum, spin and charge are all quantized.

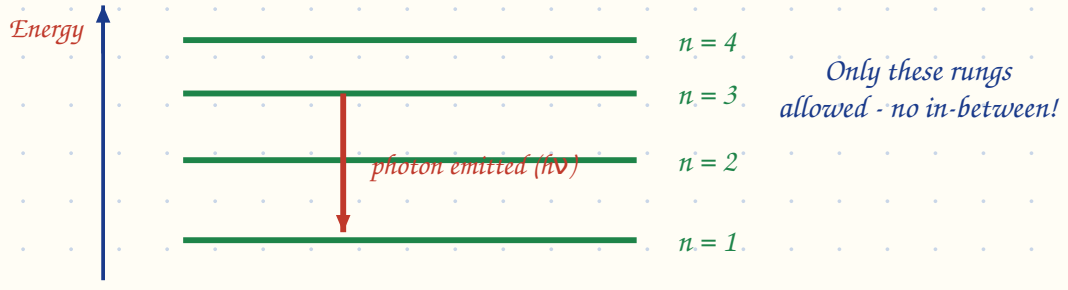


Fig 4: Discrete energy levels of an atom; a jump emits a photon

Energy levels - formulas

Hydrogen atom (Bohr): $E_n = -13.6\ eV / n^2$ ($n = 1, 2, 3 \ldots$)

Photon emitted/absorbed in a jump: $h\nu = E_{high} - E_{low}$

Particle in a 1-D box: $E_n = n^2 h^2 / 8mL^2$

Quantum harmonic oscillator: $E_n = (n + \frac{1}{2})\ \hbar\omega$ - note the non-zero ground state (zero-point energy).

- *Why discrete? Confinement of a wave forces it to fit like a standing wave - only certain wavelengths (hence energies) fit, like guitar-string harmonics.*
- *Ground state = lowest energy; excited states = higher; atoms relax by emitting a photon.*
- *Quantized levels let us build qubits: choose two levels as $|0\rangle$ and $|1\rangle$.*
- *Applications: lasers, LEDs, spectroscopy, solar cells, atomic clocks, MRI.*

Exam tip / Remember

Be ready to derive/quote: $E = h\nu$, $\lambda = h/p$, $\Delta x \Delta p \geq \hbar/2$, $E_n = -13.6/n^2\ eV$. Also know: $h = 6.626 \times 10^{-34}\ J{\cdot}s$, $1\ eV = 1.6 \times 10^{-19}\ J$.

7. Why Quantum? Strategic, Scientific & Technological Significance

» Scientific significance

- *Foundation of modern science: chemistry, materials, condensed-matter, nuclear and particle physics, cosmology all rest on quantum theory.*
- *Tests the nature of reality: non-locality, measurement, information.*
- *New simulation tool: Feynman (1981) - 'nature isn't classical, so to simulate nature use a quantum computer'.*

» Technological significance

- *First quantum revolution (1900-1980s): we learned the rules and built technologies using them - transistor, laser, LED, MRI, GPS clocks, solar cells. The digital world runs on it.*
- *Second quantum revolution (1990s-now): we now deliberately control individual quantum systems (single photons, atoms, qubits) and use superposition and entanglement as resources → quantum computers, quantum networks, quantum sensors.*

» Strategic significance

- Security: a large quantum computer running Shor's algorithm can break RSA/ECC encryption → need for post-quantum and quantum-safe cryptography ('harvest now, decrypt later' threat).
- Economy: expected to add hundreds of billions of dollars of value in pharma, finance, logistics and materials over the coming decades.
- Defence & navigation: GPS-free quantum navigation, stealth detection, secure military communication.
- Technological sovereignty: nations race to own the hardware, talent and supply chains - hence national quantum missions.

1st quantum revolution	2nd quantum revolution
Uses quantum LAWS in bulk materials	Controls INDIVIDUAL quantum states
Transistor, laser, LED, MRI, atomic clock	Qubits, QKD, quantum sensors, quantum internet
We use the effects	We engineer superposition & entanglement

8. A Snapshot of Quantum Technologies



Fig 5: The three pillars of quantum technology

» Quantum Computing

Uses qubits in superposition and entanglement to process information in fundamentally new ways. Promising for factoring (Shor), unstructured search (Grover), simulating molecules and materials, and certain optimization problems. Hardware: superconducting circuits, trapped ions, photons, neutral atoms, spins in silicon. Currently in the NISQ (noisy, intermediate-scale) era with 50 - 1000+ noisy qubits.

» Quantum Communication

Transmits quantum states (mostly photons). Main uses: Quantum Key Distribution (QKD) for provably secure keys (BB84, E91), quantum teleportation, and eventually a quantum internet linking quantum computers. Any eavesdropping disturbs the photons and is detected.

» Quantum Sensing & Metrology

Exploits the extreme sensitivity of quantum states to the environment to measure time, magnetic/electric fields, gravity, rotation and temperature with record precision. Examples: atomic clocks, NV-centre diamond magnetometers, atom interferometers (gravity gradiometers), SQUIDs for brain imaging (MEG), quantum LiDAR/radar, GPS-free inertial navigation.

Pillar	Core resource	Maturity
Computing	Superposition + entanglement	Early; prototypes with error-prone qubits
Communication	Photons, entanglement, no-cloning	QKD commercial; quantum internet = research
Sensing	Sensitivity of quantum states	Most mature; products already on market

9. National and Global Quantum Missions

Country / bloc	Programme	Highlights
India	National Quantum Mission (NQM), approved by Cabinet Apr 2023; ≈ ₹6,003 crore for 2023-24 to 2030-31; run by DST	Goals: build intermediate-scale quantum computers of 50-1000 physical qubits; satellite-based secure quantum communication over 2000 km within India; inter-city QKD over 2000 km; quantum sensing & metrology, new materials. Four Thematic Hubs (T-Hubs) at IISc Bengaluru (computing), IIT Madras (communication), IIT Bombay (sensing), IIT Delhi (materials). Startups such as QpiAI, and TIFR/ISRO/DRDO/RRI labs.
USA	National Quantum Initiative Act (2018); NQI Coordination Office; DOE & NSF quantum centres	Funding > US$1 billion/yr; players: IBM, Google, Microsoft, IonQ, Rigetti; NIST post-quantum cryptography standards (2024); Quantum-safe migration mandates.
European Union	Quantum Flagship (2018, €1 billion over 10 years); EU Quantum Strategy & Chips Act	Quantum computing, communication (EuroQCI network), sensing, simulation; strong national programmes (Germany, France, Netherlands); EuroHPC quantum computers.
China	National quantum priorities in Five-Year Plans; national lab at Hefei	Micius satellite (2016) - first satellite QKD & entanglement distribution over 1200 km; 2000+ km Beijing-Shanghai fibre backbone; Jiuzhang (photonic) and Zuchongzhi (superconducting) computers.
Others	UK National Quantum Technologies Programme; Canada, Australia, Japan, South Korea, Singapore, Finland	Combined public funding worldwide is now in the tens of billions of US dollars.

Exam tip / Remember

For India's NQM remember: 2023 - 2031, ₹6003 crore, 50-1000 qubits, 2000 km QKD, 4 T-Hubs. Compare briefly with the USA (NQI Act 2018), EU (Flagship 2018) and China (Micius satellite).

10. Solved Numerical Examples

Example 1 - Energy of a photon

Find the energy of a photon of green light, $\lambda = 500$ nm.

$E = hc/\lambda = (6.626\times10^{-34} \times 3\times10^{8}) / (500\times10^{-9}) = 3.98\times10^{-19}$ J ≈ 2.5 eV.

Example 2 - de Broglie wavelength of an electron

An electron moves at 10^6 m/s ($m = 9.1\times10^{-31}$ kg).

$\lambda = h/(mv) = 6.626\times10^{-34} / (9.1\times10^{-31} \times 10^6) = 7.3\times10^{-10}$ m ≈ 0.73 nm - similar to atomic spacing, so diffraction is observable.

Example 3 - Hydrogen transition

Electron falls from n = 3 to n = 2.

$\Delta E = -13.6(1/9 - 1/4) = 13.6 \times 0.1389 = 1.89$ eV → photon wavelength ≈ 656 nm (red H-alpha line).

Example 4 - Uncertainty principle

An electron is confined to $\Delta x = 1$ Å $= 10^{-10}$ m.

$\Delta p \geq \hbar/(2\Delta x) = 1.055\times10^{-34}/(2\times10^{-10}) = 5.3\times10^{-25}$ kg·m/s → $\Delta v \approx 5.8\times10^5$ m/s. Large! This is why electrons cannot be pinned inside atoms.

Example 5 - Probability from amplitudes

$|\psi\rangle = (\sqrt{3}/2)|0\rangle + (1/2)|1\rangle$. P(0) = 3/4 = 75%, P(1) = 1/4 = 25%. Check: 3/4 + 1/4 = 1 ✓.

11. Glossary of Key Terms

Term	Meaning
Quantum	Smallest discrete unit of a physical quantity (e.g. photon for light)
Wavefunction ψ	Mathematical object containing all information about a quantum system
Probability amplitude	Complex number whose squared magnitude gives probability
Eigenstate	State with a definite value for a measured quantity
Collapse	Sudden change of state to a definite outcome upon measurement
Interference	Adding of amplitudes giving bright/dark fringes
Planck constant ħ	6.626×10^{-34} J·s; sets the scale of quantum effects
Spin	Intrinsic angular momentum of a particle (no classical analogue)
Polarization	Orientation of oscillation of the electric field of light
Decoherence	Loss of quantum behaviour by interaction with environment
NISQ	Noisy Intermediate-Scale Quantum - today's imperfect devices
QKD	Quantum Key Distribution - secure sharing of cryptographic keys

12. Quick Revision & Likely Questions

- *Why did classical physics fail? Name 3 phenomena.*
- *State & explain the uncertainty principle with its significance.*
- *Explain superposition and entanglement with examples.*
- *Differentiate classical and quantum mechanics (table).*
- *What is quantization? Draw energy levels of hydrogen.*
- *Write a note on electrons, photons and atoms as quantum systems.*
- *Why is quantum technology strategically important? Explain 1st & 2nd quantum revolutions.*
- *Compare India's National Quantum Mission with those of the USA, EU and China.*

One-line summary

Quantum theory → discrete energies, wave-particle duality, probabilistic measurement → superposition & entanglement → new technologies in computing, communication and sensing → global race for leadership.

UNIT 2

Theoretical Structure of Quantum Information Systems

Unit 2 at a glance

Qubit (spin & polarization) · bits vs qubits · physical systems (trapped ions, superconducting, photons) · coherence & decoherence · Hilbert space, states, operators (abstract) · entanglement & non-locality · quantum vs classical information · philosophical implications

1. What is a Qubit?

Classical information is stored in a bit, which is always either 0 or 1. Quantum information is stored in a qubit (quantum bit): any two-level quantum system whose two basis states are labelled $|0\rangle$ and $|1\rangle$. Unlike a bit, a qubit can be in a superposition of both:

Qubit state

$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$

α, β = complex probability amplitudes. $P(0) = |\alpha|^2$, $P(1) = |\beta|^2$.

Equal superposition: $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ → 50 % chance of each result.

When we measure a qubit we never see 'a bit of both' - we get a single classical bit, 0 or 1, with the above probabilities. Before measurement the amplitudes behave like waves and can interfere. The art of quantum computing is to make wrong answers cancel and right answers reinforce.

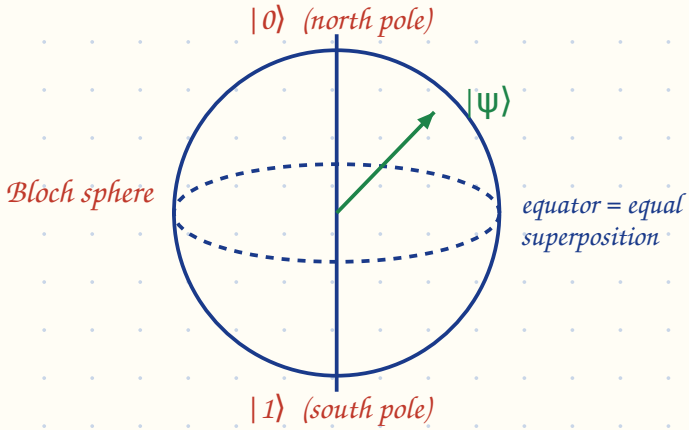


Fig 1: Bloch sphere - geometric picture of one qubit

» Bloch sphere picture

- Every pure single-qubit state is a point on the surface of a sphere of radius 1.
- North pole = $|0\rangle$, south pole = $|1\rangle$; points on the equator are equal superpositions with different relative phase.
- $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$ - θ is the latitude angle, φ the longitude (phase).

- Quantum gates = rotations of the arrow on this sphere. Noise = arrow shrinking/drifting.

2. Physical Realizations: Spin and Polarization



Fig 2: Two natural two-level systems

» 2.1 Spin qubit (electron or nucleus)

- Spin is an intrinsic angular momentum, like a tiny magnet. For spin-½, measurement along an axis gives only two results: up (↑) or down (↓).
- Map: $|\uparrow\rangle \to |0\rangle$, $|\downarrow\rangle \to |1\rangle$. A magnetic field splits the two energy levels; microwave/RF pulses rotate the spin.
- Measuring along z gives ↑/↓; measuring along x gives left/right - 'incompatible' directions: an up state is a 50-50 superposition of left & right (uncertainty principle at work).
- Realizations: electrons in quantum dots, phosphorus in silicon, NV centres in diamond, nuclear spins (NMR).

» 2.2 Polarization qubit (photon)

- Polarization = direction of oscillation of light's electric field. Choose H (horizontal) = $|0\rangle$, V (vertical) = $|1\rangle$.
- Diagonal $D = (|H\rangle + |V\rangle)/\sqrt{2}$, anti-diagonal $A = (|H\rangle - |V\rangle)/\sqrt{2}$; circular R/L also possible.
- Manipulated with wave plates ($\lambda/2$, $\lambda/4$), polarizing beam splitters; measured by polarizers + single-photon detectors.
- Test: a polarizer at angle θ passes a photon with probability $\cos^2\theta$ (Malus's law at single-photon level).

> **Exam tip / Remember**
> Spin and polarization are mathematically the same: a two-dimensional state space. Learn this analogy - examiners love it: spin up/down ↔ polarization H/V ↔ $|0\rangle/|1\rangle$.

3. Classical Bits vs Quantum Bits



Fig 3: A bit has 2 states; a qubit has infinitely many

Property	Classical bit	Qubit
Possible states	0 or 1	$\alpha \|0\rangle + \beta \|1\rangle$ (continuum)
Physical carrier	Voltage, magnetic domain, transistor	Spin, photon, ion, superconducting circuit
Reading it	Does not disturb the value	Measurement collapses state; gives 0/1 randomly
Copying	Easy (fan-out)	Impossible for unknown states (no-cloning)
n units describe	1 value out of 2^n at a time	2^n amplitudes at once (superposition)
Correlation	Classical (shared randomness)	Entanglement - stronger than classical
Error behaviour	Digital: robust, simple error correction	Analog-like errors; needs quantum error correction
Operations	AND, OR, NOT (irreversible)	Unitary gates (reversible) + measurement
Stability	Stable for years	Fragile: microseconds to seconds

» Growth of information: why n qubits are special

n classical bits hold ONE of 2^n strings. n qubits can be in a superposition of ALL 2^n strings with an amplitude for each. 10 qubits → 1024 amplitudes; 50 qubits → about 10^{15}; 300 qubits → more numbers than atoms in the universe. A classical computer would need exponentially large memory to store such a state. But beware: measurement returns only n bits, so extracting useful results needs clever algorithms.

★ *Common myth*

A quantum computer does NOT simply try all answers at once and read the best one. Interference must be designed so the right answer is likely at measurement.

4. Physical Quantum Systems for Qubits (Non-engineering View)

Platform	What is the qubit?	How controlled	Good / Bad
Superconducting circuits	Two lowest energy states of an electrical circuit (transmon) cooled to ~15 mK; behaves like an 'artificial atom'	Microwave pulses; resonators for read-out	Fast gates (ns), chip-based, scalable / short coherence (~100 μs), needs dilution refrigerator
Trapped ions	Two internal energy levels of a single charged atom (e.g. Yb^+, Ca^+) held in vacuum by electric fields	Laser pulses; ions share motion to interact	Highest fidelity, long coherence (seconds-minutes), all-to-all links / slow gates, scaling to large ion chains is hard
Photons	Polarization, path or time-bin of a single particle of light	Beam splitters, phase shifters, detectors	Room temperature, travels far, ideal for communication / photons don't interact easily, losses

Other platforms: neutral atoms in optical tweezers, spins in silicon/quantum dots, NV centres in diamond, topological (Majorana, still experimental). All satisfy the same abstract qubit rules - the platform differs only in how well it can be controlled.

5. Quantum Coherence and Decoherence (Intuitive)

Coherence means the superposition keeps a well-defined phase relationship between its parts - like two perfectly synchronized waves - so interference is possible. Decoherence is the loss of that phase relationship because the qubit unintentionally interacts (exchanges energy/information) with its environment: stray photons, heat, magnetic noise, vibrations, neighbouring qubits.

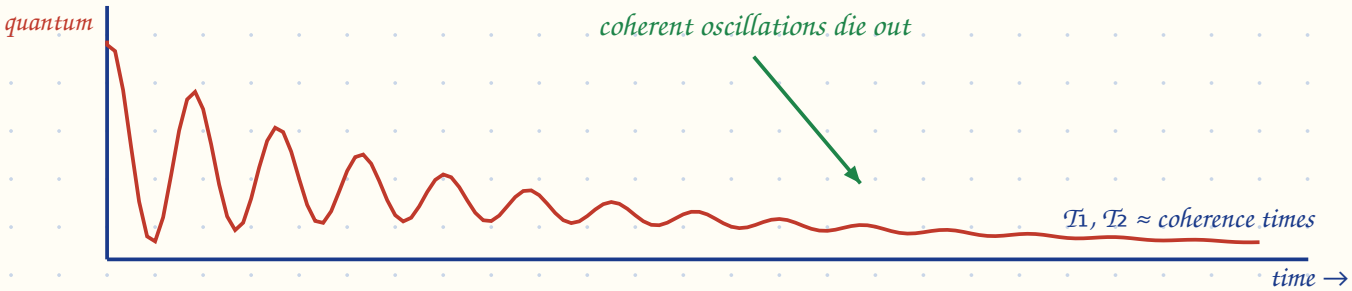


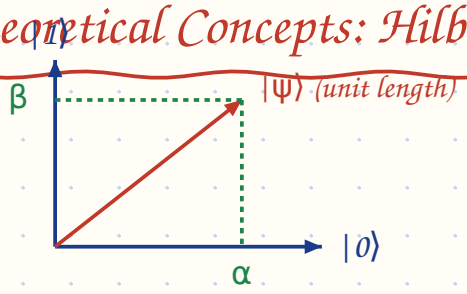
Fig 4: Quantum behaviour fades with time as the environment 'listens'

- Intuition: imagine a spinning coin balanced on its edge. Any slight bump makes it fall - the environment 'measures' it. The more particles it touches, the faster it falls.
- T_1 (relaxation time): time for $|1\rangle$ to decay into $|0\rangle$ by losing energy.
- T_2 (dephasing time): time for the phase between $|0\rangle$ and $|1\rangle$ to be randomized; always $T_2 \leq 2T_1$.

- *Gate operations must finish ≪ T_2. The figure of merit is (coherence time) ÷ (gate time) = number of operations possible.*
- *Bigger systems decohere faster - explaining why the everyday world looks classical (Zurek's decoherence theory).*

> *Fighting decoherence*
> *Isolation (vacuum, shielding, cryogenics) · better materials · dynamical decoupling (echo pulses) · decoherence-free subspaces · quantum error correction.*

6. Theoretical Concepts: Hilbert Space, States, Operators (Abstract)



state = vector
operator = matrix/rotation

measurement = project
then read length²

Fig 5: A state is a unit vector in an abstract space

» 6.1 Hilbert space

A Hilbert space is the abstract mathematical 'stage' for a quantum system: a complex vector space with an inner product (a way to measure lengths and angles/overlaps). Each physical system has its own Hilbert space; a qubit lives in a 2-dimensional one ($\mathbb{C}^2$). Think of it as the set of all allowed 'directions' the system's state can point.

» 6.2 Quantum states

- *A pure state = a unit-length vector $|\psi\rangle$ in the space (rays differing by an overall phase are physically identical).*
- *The inner product $\langle\varphi|\psi\rangle$ is the 'overlap amplitude'; $|\langle\varphi|\psi\rangle|^2$ = probability of finding φ if the system is in ψ.*
- *Basis states: a set of mutually perpendicular unit vectors ($|0\rangle$, $|1\rangle$). Any state = combination of them.*
- *A mixed state (density matrix ρ) = statistical mixture of pure states - describes ignorance or entanglement with something else.*

» 6.3 Operators

- *Operators act on states: $\hat{A}|\psi\rangle = |\psi'\rangle$. In practice they are matrices.*

- *Unitary operators* ($U^\dagger U = I$) = reversible rotations preserving length: all quantum gates and time evolution.
- *Hermitian operators* = observables (energy, spin, position): their eigenvalues are possible measurement results; eigenvectors are the states with those definite values.
- Non-commuting operators ($AB \neq BA$) cannot be measured sharply together → uncertainty principle.

Key operators (just remember the roles)
X = bit-flip: $|0\rangle \leftrightarrow |1\rangle$ · Z = phase-flip: $|1\rangle \to -|1\rangle$ · H (Hadamard) = makes superposition: $|0\rangle \to (|0\rangle + |1\rangle)/\sqrt{2}$.
Tensor product $\otimes$ combines systems: 2 qubits live in a 4-dimensional space $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

7. The Role of Entanglement and Non-locality

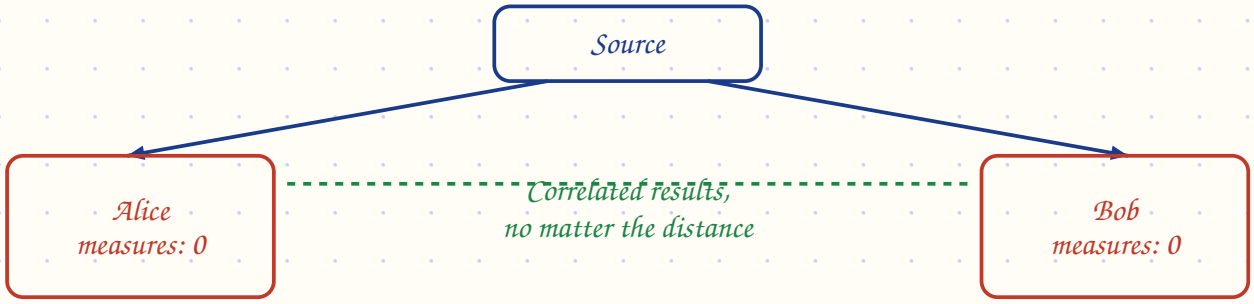


Fig 6: Entangled pair - perfectly correlated outcomes

Two qubits are *entangled* if their joint state cannot be written as a product of separate states. The classic example is the Bell state:

Bell state
$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$
Measure qubit 1: get 0 or 1 with 50 % each. Then qubit 2 is always found with the SAME result. Yet neither qubit alone has a definite value before measurement.

- *Non-locality*: Bell's inequality tests show correlations stronger than any local-hidden-variable theory can give (CHSH bound 2 vs quantum $2\sqrt{2} \approx 2.83$). Loophole-free experiments (2015) confirmed this.
- No faster-than-light signalling - Alice's outcomes alone look random; the correlation shows up only when results are compared via ordinary communication.
- Role in quantum information: *teleportation* (transfer a state using entanglement + 2 classical bits), *superdense coding* (send 2 classical bits with 1 qubit + entanglement), QKD (E91), and the speed-up of quantum algorithms (entanglement lets qubits' states depend on each other).

Monogamy of entanglement

If two qubits are maximally entangled, neither can be entangled with a third. This is why eavesdroppers are detectable in entanglement-based cryptography.

8. Quantum Information vs Classical Information

Aspect	Classical information	Quantum information
Unit	bit (0/1)	qubit; also qudit, continuous variables
Copying	Freely copied, broadcast	No-cloning theorem forbids copying unknown states
Reading	Non-destructive, repeated	Measurement collapses; one copy gives ≤ 1 bit (Holevo bound)
Deleting / hiding	Can be deleted irreversibly	No-deleting theorem; evolution is reversible
Correlations	Local; shared randomness	Entanglement, non-local
Teleporting	Meaningless - just copy	Teleportation: moves the state; original destroyed
Security	Computational assumptions (RSA)	Laws of physics (QKD)
Theory	Shannon entropy $H = -\Sigma p \log p$	von Neumann entropy $S = -Tr\ \rho \log \rho$
Errors	Bit flips only	Bit flips, phase flips, continuous drifts

Three golden theorems

No-cloning: unknown quantum state cannot be copied. No-deleting: unknown state cannot be erased given a copy. No-broadcasting: cannot send the state to many users while keeping it intact. Together they make quantum information special and secure.

9. Philosophical Implications

» Randomness vs determinism

Classical physics (Laplace's demon) is deterministic: know the present exactly and you can compute the future. In quantum mechanics the state evolves deterministically (Schrödinger equation) but individual measurement outcomes are fundamentally random - only probabilities are lawful. Quantum random number generators exploit this to produce genuine, not pseudo-, randomness.

» The role of the observer

- Measurement links the quantum and the classical - the 'measurement problem': why does the state collapse to ONE outcome?
- Copenhagen view: wavefunction = knowledge; collapse upon measurement; the cut between system and apparatus is pragmatic.
- Many-Worlds (Everett): no collapse; all outcomes occur in branching universes.
- Pilot-wave (de Broglie-Bohm): deterministic hidden variables, but non-local.
- QBism / information-based views: quantum states are an agent's degrees of belief.
- Decoherence explains why interference vanishes but not, by itself, why we see one outcome.

» Realism and locality

Einstein-Podolsky-Rosen (1935) asked if quantum theory is complete. Bell (1964) showed that no theory that is both local and realistic can match quantum predictions. So we must give up locality, realism, or both - a deep philosophical shift. Yet the rules are consistent and extremely well tested.

> **Exam tip / Remember**
> For essay questions: present (i) determinism vs randomness, (ii) the measurement/observer problem with 2-3 interpretations, (iii) non-locality (EPR/Bell), (iv) practical payoff - randomness and security used in technology.

9A. Measurement Bases and Basis Dependence

A qubit can be measured in different bases. The state $|0\rangle$ is certain in the Z-basis $\{|0\rangle, |1\rangle\}$ but is a coin-toss in the X-basis $\{|+\rangle, |-\rangle\}$. So information about 'which basis' matters - the foundation of BB84 cryptography.

Basis	States	Spin picture	Photon picture
Z (computational)	$\|0\rangle$, $\|1\rangle$	up / down	H / V (0° / 90°)
X (Hadamard)	$\|+\rangle$, $\|-\rangle$	right / left	D / A (+45° / −45°)
Y (circular)	$\|R\rangle$, $\|L\rangle$	in / out	right / left circular

- Measuring in the wrong basis gives random results and disturbs the state.
- Complementary (mutually unbiased) bases: a state definite in one is maximally uncertain in the other.
- Rule of thumb: the more sure you are about the Z-value, the less you know about X-value - uncertainty principle again.

9B. Pure vs Mixed States (Density Matrix - Idea Only)

Pure state	Mixed state
Complete knowledge: a single vector $\|\psi\rangle$	Partial knowledge: a probabilistic mixture of states
Point on surface of Bloch sphere	Point inside the sphere (centre = completely random)
Superposition: coherent, shows interference	Mixture: classical ignorance, no interference
Example: $\|+\rangle = (\|0\rangle + \|1\rangle)/\sqrt{2}$	Example: 50 % $\|0\rangle$ and 50 % $\|1\rangle$ (a noisy coin)

Important subtlety: a half of an entangled pair, looked at alone, is a maximally mixed state even though the full pair is in a pure state. Decoherence is exactly this process - the qubit becomes entangled with the environment, so on its own it looks like a mixture.

9C. Two Protocols that Show Quantum Information at Work

» Quantum teleportation (steps)

1. Alice and Bob share an entangled pair (one qubit each).
2. Alice has an unknown qubit $|\psi\rangle$ to send. She performs a joint (Bell) measurement on $|\psi\rangle$ and her half of the pair.
3. She gets 2 classical bits (00, 01, 10 or 11) and tells Bob by ordinary communication.
4. Depending on the bits, Bob applies I, X, Z or XZ to his qubit - it now equals $|\psi\rangle$.
5. Original $|\psi\rangle$ is destroyed (consistent with no-cloning). No matter has moved, no faster-than-light signal.

» Superdense coding

With a shared entangled pair, Alice can send two classical bits by transmitting only one qubit: she applies I, X, Z or XZ to her half and sends it; Bob does a Bell measurement on both halves and reads 2 bits. Teleportation and superdense coding are 'mirror images' of each other.

> Resources table
>
> Teleportation: 1 ebit + 2 cbits → 1 qubit. Superdense coding: 1 ebit + 1 qubit → 2 cbits. (ebit = shared entangled pair, cbit = classical bit)

9D. Formula & Concept Sheet

Concept	Formula / statement
Qubit	$\|\psi\rangle = \alpha\|0\rangle + \beta\|1\rangle$, $\|\alpha\|^2 + \|\beta\|^2 = 1$
Bloch form	$\cos(\theta/2)\|0\rangle + e^{i\varphi}\sin(\theta/2)\|1\rangle$
Born rule	$P(i) = \|\langle i\|\psi\rangle\|^2$
Hadamard	$H\|0\rangle = \|+\rangle$, $H\|1\rangle = \|-\rangle$
Two-qubit space	dimension $2^2 = 4$; n qubits $\rightarrow 2^n$
Bell state	$(\|00\rangle + \|11\rangle)/\sqrt{2}$
Unitary	$U^\dagger U = I$ (reversible)
CHSH	Classical ≤ 2; Quantum $\leq 2\sqrt{2}$
Coherence	$T_2 \leq 2T_1$
No-cloning	No U with $U\|\psi\rangle\|0\rangle = \|\psi\rangle\|\psi\rangle$ for all ψ

10. Solved Examples

Example 1

Qubit $|\psi\rangle = (1/\sqrt{5})|0\rangle + (2/\sqrt{5})|1\rangle$. $P(0) = 1/5 = 20\ \%$, $P(1) = 4/5 = 80\ \%$. Normalized since $1/5 + 4/5 = 1$.

Example 2

How many complex numbers describe a general 3-qubit state? $2^3 = 8$ amplitudes (minus 1 for normalization and 1 for global phase $\Rightarrow$ 14 real parameters).

Example 3

Apply Hadamard to $|1\rangle$: $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2} = |-\rangle$. Measurement probabilities still 50/50, but the sign (phase) differs - detectable by a second H, which returns $|1\rangle$ with certainty. Shows interference!

Example 4

Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$: $P(00) = ½$, $P(11) = ½$, $P(01) = P(10) = 0$ - outcomes always agree.

11. Revision Questions

1. Define a qubit. Explain with spin and polarization.

2. Compare classical bits and qubits in a table.
3. Describe trapped-ion, superconducting and photonic qubits.
4. What are coherence and decoherence? Define T_1 and T_2.
5. Explain Hilbert space, states and operators in abstract terms.
6. What is entanglement? Explain non-locality and its role in quantum information.
7. State the no-cloning theorem and its importance.
8. Discuss philosophical implications: randomness, determinism, observer.

12. Short Answer Practice (Model Answers)

Q1. Why can't we clone a qubit?

Quantum evolution is linear (unitary). A cloning machine that copies $|0\rangle$ and $|1\rangle$ would, by linearity, map $(|0\rangle + |1\rangle)/\sqrt{2}$ to an entangled state, not to two copies. Hence no device can copy an arbitrary unknown state. Consequence: eavesdroppers cannot copy photons in QKD.

Q2. Why is a qubit more than a probabilistic bit?

A probabilistic classical bit is a mixture with real probabilities p, 1−p. A qubit has complex amplitudes with phases, so contributions can cancel (destructive interference) - something probabilities never do. Also several qubits can be entangled.

Q3. Why must quantum gates be unitary?

Unitarity preserves total probability (norm = 1) and makes operations reversible. Any non-unitary change (loss, noise) is treated as an error.

Q4. Does entanglement allow faster-than-light communication?

No. Each party's outcomes are random; correlations are visible only after comparing results over a classical channel limited by light speed.

Q5. Which qubit platform is best?

No single winner. Superconducting: fast and chip-friendly. Trapped ions: highest accuracy and connectivity. Photons: best for communication and room temperature. Choice depends on the task and the trade-off between speed, coherence and scalability.

Q6. What is the measurement problem?

Quantum theory has two rules - smooth unitary evolution and abrupt collapse on measurement. Why/when collapse occurs and what counts as a measurement is unresolved; interpretations (Copenhagen, Many-Worlds, Bohm, QBism) offer different answers while predicting the same numbers.

★ *Final summary*

Qubit = two-level system with complex amplitudes → lives in Hilbert space → evolves by unitary operators → fragile to decoherence → entanglement links qubits non-locally → quantum information obeys no-cloning, reversibility and collapse rules → deep questions of randomness and observer.

UNIT 3

Building a Quantum Computer - Theoretical Challenges & Requirements

Unit 3 at a glance

What is needed · fragility: decoherence, noise, control · conditions: isolation, error management, scalability, stability · why entanglement is hard to keep · error correction · hardware: superconducting, trapped ions, photonics · vision vs reality · role of quantum software

1. What is Required to Build a Quantum Computer?

A quantum computer is not a faster laptop. It is a machine that stores information in qubits, manipulates them with precisely timed physical operations, and reads out a result. David DiVincenzo (2000) listed the minimum requirements - now called the DiVincenzo criteria.

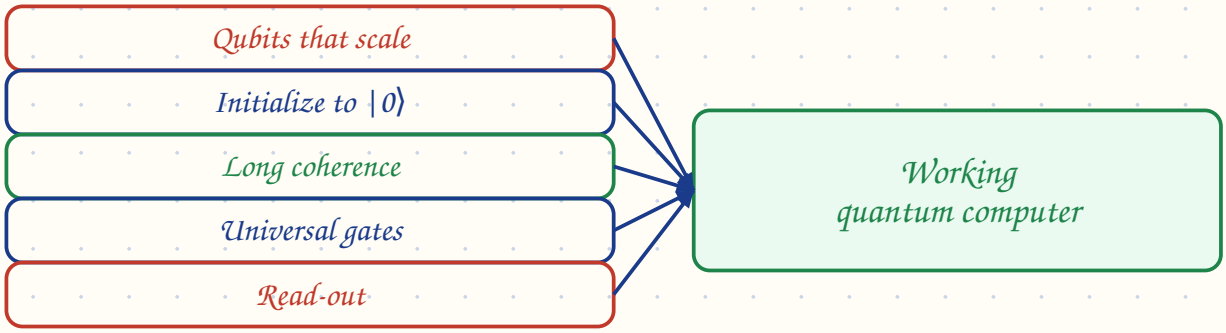


Fig 1: The DiVincenzo checklist

1. *Scalable physical system with well-defined qubits - many identical, well-characterized two-level systems.*
2. *Initialization - ability to reset all qubits to a known state such as $|000\ldots0\rangle$.*
3. *Long coherence times - far longer than the time of one gate operation.*
4. *Universal set of quantum gates - e.g. single-qubit rotations + one entangling two-qubit gate (CNOT/CZ).*
5. *Qubit-specific measurement - read out each qubit reliably.*
6. *(For networking) interconvert stationary and flying qubits; transmit them faithfully.*

★ *In one line*

Build good qubits, keep them quiet, control them precisely, read them out, and do it for millions of them.

» Ingredients in practice

Layer	What is needed
Qubits	Two-level quantum systems (circuits, ions, photons, spins)
Environment	Ultra-high vacuum or ~10-20 millikelvin refrigeration, magnetic shielding, vibration isolation
Control	Precisely shaped microwave / laser / voltage pulses with nanosecond timing
Read-out	Amplifiers, single-photon detectors, fast electronics
Classical computer	Real-time decoding, calibration, pulse sequencing
Software	Compilers, error-correction decoders, algorithms

2. Fragility of Quantum Systems

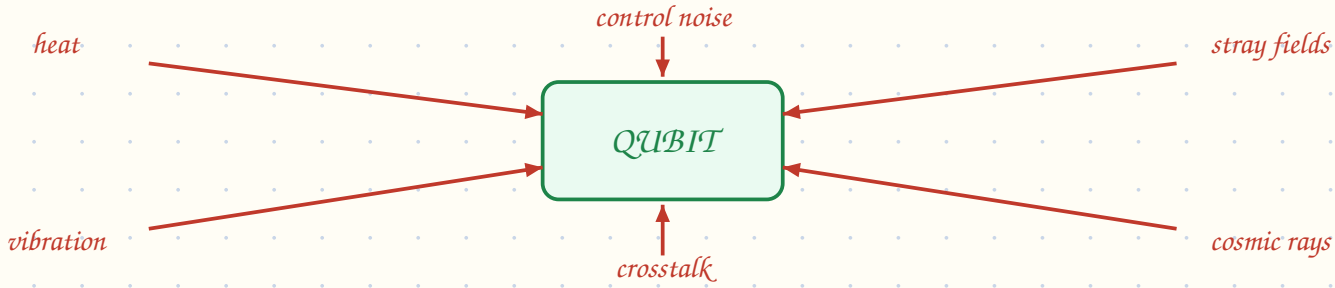


Fig 2: Everything in the environment attacks the qubit

» 2.1 Decoherence

The tiniest unwanted interaction acts like an unintended measurement and destroys superposition (see Unit 2). Sources: thermal vibrations, stray electromagnetic fields, defects in materials, cosmic rays, nearby qubits.

» 2.2 Noise

- Bit-flip noise: $|0\rangle \leftrightarrow |1\rangle$ by accident.
- Phase-flip noise: relative sign changes ($|+\rangle \leftrightarrow |-\rangle$) - unique to quantum, no classical counterpart.
- Amplitude damping: $|1\rangle$ loses energy and falls to $|0\rangle$ (T_1).
- Leakage: qubit escapes to a higher energy level outside the two levels.
- Crosstalk: control pulse for one qubit leaks to neighbours.

» 2.3 Control errors

Gates are analog operations: a rotation by an angle that is slightly off, or a pulse of slightly wrong length, produces a small error. Errors are continuous, accumulate over many gates and cannot be fixed by 'looking' (measurement would collapse the state).

Typical numbers (approximate, platform dependent)

Superconducting: gate ~20-100 ns, T_2 ~ 50-300 μs, 2-qubit error ~0.1-1 %. Trapped ion: gate ~ 10-100 μs, T_2 ~ seconds, 2-qubit error ~0.05-0.5 %.

A 1000-gate algorithm with 1 % error per gate has success probability ~ $(0.99)^{1000} \approx 0.004$ % - essentially zero. Hence error correction.

3. Conditions for a Functional Quantum System

Condition	Meaning	How achieved
Isolation	Protect qubits from environment but still allow control and read-out - a paradox: perfectly isolated qubits cannot be used	Vacuum, cryostats (10 mK), shielding, careful materials, suspended ions
Error management	Detect and correct errors faster than they accumulate	Quantum error correction codes, error mitigation, calibration
Scalability	Go from tens to millions of qubits without losing quality	Modular chips, interconnects, multiplexed control, 3-D wiring, cryo-CMOS
Stability	Keep device behaviour steady over hours/days	Automatic recalibration, temperature control, drift tracking

» Trade-offs - the 'quantum paradox'

- Want isolation (to stay quantum) vs want strong control (to compute).
- More qubits → more connections → more noise paths → lower quality per qubit.
- Faster gates give less decoherence per gate but need more powerful, noisier control.
- Cooling to millikelvin requires dilution refrigerators with limited wiring space - the 'wiring bottleneck'.

Exam tip / Remember

Remember the four conditions as I-E-S-S: Isolation, Error management, Scalability, Stability.

4. Theoretical Barriers: Why Maintaining Entanglement Is Difficult

- Entanglement is a collective property: the more qubits entangled, the more ways an environmental disturbance on any one of them ruins the whole state.
- Decoherence rate grows with system size - roughly linear in the number of entangled qubits (and even faster for 'cat-like' states, e.g. GHZ states).

- *Monogamy of entanglement: entanglement with the environment steals entanglement from your qubits.*
- *Entanglement is fragile to loss: for photons, a lost photon removes the entanglement entirely; fibre loss is exponential with distance (~0.2 dB/km).*
- *Gate imperfections: each entangling gate has the highest error rate (2-qubit errors ≫ 1-qubit errors).*
- *Long-range entanglement needs quantum repeaters because states can't be cloned or amplified.*

> *Entanglement lifetime picture*
> *If one qubit has coherence time T, N entangled qubits typically behave like they have T/N. A 100-qubit entangled state may last only 1/100 as long as one qubit!*

5. Error Correction as a Theoretical Necessity

Classical computers fix errors by redundancy: store 0 as 000, 1 as 111 and use majority vote. In quantum computing we cannot copy qubits (no-cloning) or look at them directly (collapse). Yet Peter Shor (1995) and Steane (1996) showed that quantum error correction (QEC) is possible!

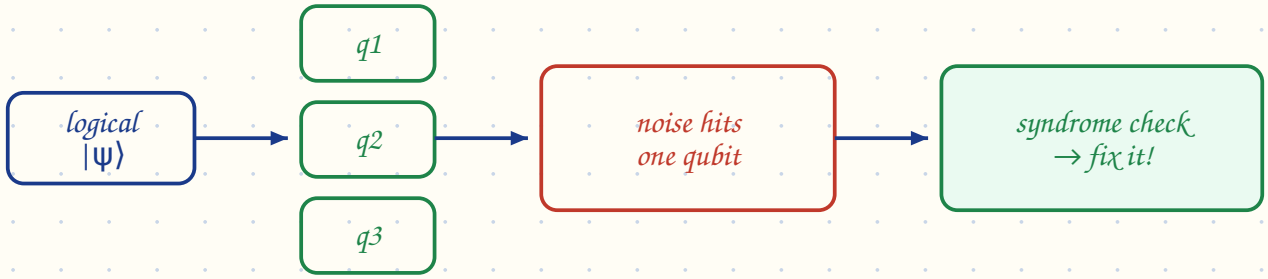


Fig 3: Idea of a quantum error-correcting code

» The key tricks

1. *Encode one logical qubit into many physical qubits using entanglement - not copying:* $\alpha|0\rangle+\beta|1\rangle \rightarrow \alpha|000\rangle+\beta|111\rangle$.
2. *Measure syndromes: parity checks on pairs of qubits (e.g. 'are q1 and q2 equal?') reveal WHERE an error happened without revealing* α, β.
3. *Correct by applying a fix gate to the flagged qubit.*
4. *Phase errors are handled by an additional code in the X-basis (Shor's 9-qubit code handles both).*

Threshold theorem

If the error rate per physical gate is below a critical value (the threshold, ~1 % for surface code), then by using more and more physical qubits the logical error can be made arbitrarily small. → Fault-tolerant quantum computing is possible in principle.

Code	Idea	Note
3-qubit repetition	Majority vote against bit flips	Teaching example
Shor 9-qubit	Handles any single-qubit error	First QEC code (1995)
Steane 7-qubit	CSS code from classical Hamming code	Good transversal gates
Surface code	2-D grid, local parity checks	Leading candidate; ~1000 physical per logical qubit
qLDPC / bosonic (cat, GKP)	More efficient encoding	Active research (IBM, Google, AWS)

- Overhead: a useful algorithm (e.g. RSA-2048 factoring) may need ~10^3-10^4 logical qubits → millions of physical qubits.
- Error mitigation (extrapolation, probabilistic cancellation) is used on NISQ devices without full correction - useful but not scalable.
- Milestone: Google (2023-24) showed logical error falling as code size grows (below threshold).

6. Quantum Hardware Platforms - Brief Conceptual Comparison

» 6.1 Superconducting circuits

Tiny electrical circuits made of aluminium/niobium on silicon, cooled below 20 mK where they become superconducting. A Josephson junction makes the circuit's energy levels uneven (anharmonic), so the lowest two levels act as a qubit (transmon). Used by IBM, Google, Rigetti, IQM, Alibaba, Fujitsu/RIKEN. Strengths: fast gates, lithographic fabrication like chips. Weaknesses: short coherence, fabrication variability, cryogenic wiring.

» 6.2 Trapped ions

Individual atoms (Yb^+, Ca^+, Ba^+) held in vacuum by oscillating electric fields; qubit = two internal levels; manipulated with lasers; entangled via shared vibrational motion. Used by IonQ, Quantinuum, Alpine Quantum Technologies. Strengths: identical qubits by nature, best gate fidelities (>99.9 %), all-to-all connectivity. Weaknesses: slower gates, scaling lasers & large traps (modular 'QCCD' architectures help).

» 6.3 Photonics

Qubits are photons (path/polarization/time-bin); gates via beam splitters, interferometers and detectors on silicon-nitride chips. Used by PsiQuantum, Xanadu, Quandela, USTC. Strengths: room-temperature operation (detectors may be cooled), natural for networking, low decoherence. Weaknesses: photon loss, probabilistic two-photon gates, need for many sources/detectors, large overhead.

Feature	Superconducting	Trapped ions	Photonics
Qubit	Circuit energy levels	Atomic levels	Photon modes
Temperature	~10 mK	Vacuum; ions laser-cooled; chamber ~room-T or 4 K	Room temp (detectors ~ 2-4 K)
Gate speed	Very fast (ns)	Slow (μs-ms)	Very fast (ps) but probabilistic
Coherence	~0.1 ms	seconds-minutes	limited by loss, not time
2-qubit fidelity	~99.5-99.9 %	~99.9 %+	depends on photon quality
Connectivity	Nearest neighbour (fixed)	All-to-all (in chain)	Flexible via waveguides
Scalability	Fab-friendly; wiring bottleneck	Trap/laser scaling	Manufacturable; loss management
Players	IBM, Google, Rigetti, IQM	IonQ, Quantinuum	PsiQuantum, Xanadu, Quandela

- Other: neutral atoms (QuEra, Pasqal, Atom Computing - 1000+ qubits arrays), silicon spin qubits (Intel, Diraq), NV centres, topological qubits (Microsoft's Majorana effort).
- Different platforms may win for different tasks - computing vs networking vs sensing; hybrid designs likely.

7. Vision vs Reality: What Works and What's Elusive

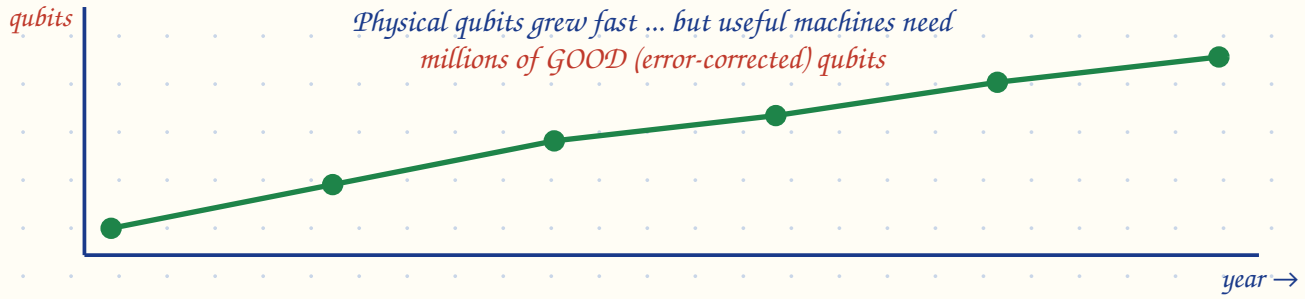


Fig 4: Qubit counts have exploded, but quality & error correction lag

Working today	Still elusive
Tens to ~1000+ noisy physical qubits (IBM Condor 1121; Atom Computing ~1200)	Large-scale fault-tolerant machines with 10^3-10^4 logical qubits

Working today	Still elusive
Quantum supremacy / advantage demos on contrived tasks (Google 2019 & 2024, USTC Jiuzhang)	Clear, practical advantage on useful commercial problems
Error-corrected logical qubits in small demonstrations (Google Willow, Quantinuum, QuEra/Harvard)	Reducing overhead of error correction (millions of qubits)
Cloud access to real devices (IBM Quantum, Amazon Braket, Azure Quantum)	Cheap, compact, reliable cryogenics & control wiring
QKD networks, quantum sensors, quantum random number generators	Quantum memory, repeaters and a global quantum internet
Algorithms known: Shor, Grover, quantum simulation, QAOA, VQE	More algorithms with clear exponential benefits

Honest summary

Hardware progress is real and accelerating, but a 'cryptographically relevant' or 'broadly useful' quantum computer is still years away (estimates range ~5 to 20 years). Hype must be balanced with realism - the 'quantum winter' risk if expectations are not met.

8. The Role of Quantum Software

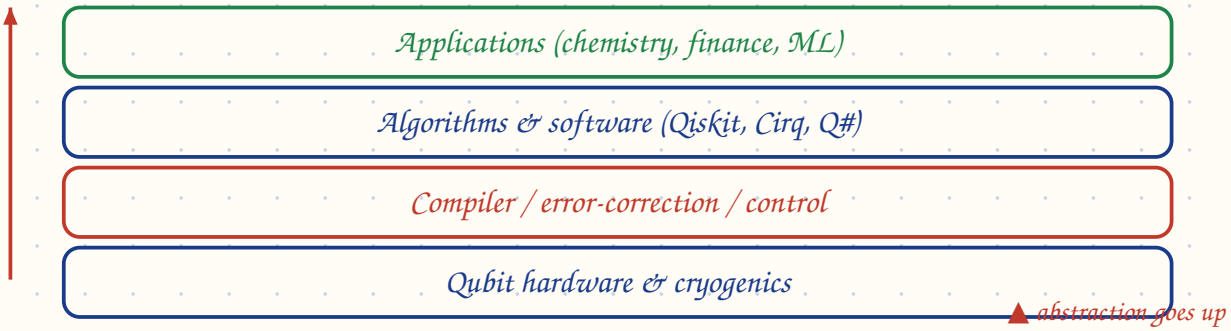


Fig 5: The quantum computing stack

Quantum software hides the physics and manages complexity so that algorithm designers need not control microwave pulses directly.

- *Programming frameworks*: IBM Qiskit, Google Cirq, Microsoft Q# / Azure Quantum, Xanadu PennyLane, Amazon Braket SDK, Quantum Development Kits.
- *Compilation / transpilation*: translate abstract circuits into the device's native gates, map logical qubits to physical ones respecting connectivity, minimize gates & depth.
- *Error mitigation & QEC software*: noise-aware compilation, zero-noise extrapolation, decoders that process syndrome data in real time.
- *Calibration & control software*: auto-tune pulses, track drift, schedule experiments.

- *Simulators & emulators: classical simulation of ~30-50 qubits for testing and education; noise models.*
- *Resource estimation: estimate the physical qubits and time needed for a target algorithm.*
- *Hybrid quantum-classical workflows: variational algorithms (VQE, QAOA) in which a classical optimizer tunes a quantum circuit; HPC integration.*

8A. Engineering-Free View of the Hardware Environment

» Why so cold? (superconducting & spin qubits)

Thermal energy kT must be much smaller than the qubit's energy gap hf. A transmon with $f \approx 5$ GHz has $hf/k \approx 0.24$ K, so we need $T \approx 10$-20 mK - colder than outer space (2.7 K). A dilution refrigerator reaches this using a mixture of helium-3 and helium-4, with stages at 50 K → 4 K → 800 mK → 100 mK → 10 mK. The 'chandelier' of gold-plated plates and coaxial cables is the iconic picture of a quantum computer.

» Why vacuum and lasers? (trapped ions, neutral atoms)

A single stray gas molecule colliding with an ion destroys its state, so ultra-high vacuum (~10^{-11} mbar) is used. Lasers cool the ions close to absolute zero (Doppler/sideband cooling), prepare them in a known state, drive gates, and make them fluoresce for read-out (bright = 1, dark = 0).

» Why precise light? (photonics)

Photon sources must emit one photon at a time, indistinguishable from each other; waveguides must have very low loss; detectors (superconducting nanowires) must be efficient. Because photons do not interact with each other, two-photon gates are built using measurement and ancillary photons ('measurement-based' or 'fusion-based' computing).

Type of noise	Cause	Remedy
Thermal	Not cold enough; hot wires	Better refrigeration, attenuation & filtering of lines
Magnetic/charge fluctuations	Material defects, two-level fluctuators	Cleaner fabrication, different qubit geometry (transmon), spin echo
Laser noise	Intensity/phase/frequency jitter	Stabilized lasers, composite pulses
Photon loss	Absorption, scattering, coupling losses	Better materials, heralded schemes, redundancy
Cosmic rays	High-energy particles break Cooper pairs → correlated errors	Shielding, underground labs, qubit design
Control electronics	Timing jitter, finite precision	High-quality DACs, calibration, cryo-CMOS

8B. Surface Code in Brief (Intuitive)

The surface code arranges data qubits on a checkerboard grid with measure qubits between them. Measure qubits repeatedly check 'parities' of their four neighbours. When an error occurs, some checks flip (a 'defect' appears); a classical decoder looks at the pattern of defects over time to guess the most probable error chain and correct it.

- *Needs only nearest-neighbour interactions on a 2-D chip - hardware friendly.*
- *Threshold ≈ 1 % (physical error rate must be below this).*
- *Larger code distance d → logical error falls roughly as $(p/p_th)^{(d+1)/2}$.*
- *Cost: d^2 data qubits per logical qubit (≈ $2d^2$ including measure qubits).*
- *Logical operations by 'lattice surgery' (merging/splitting code patches) or braiding defects.*

> *Fault-tolerance: three golden rules*
>
> *1. Error correction must not introduce more errors than it removes (below threshold). 2. A single fault must not spread to multiple qubits in one block. 3. Fresh, clean ancilla qubits are needed continuously - requires fast reset and measurement.*

8C. Key Challenges at a Glance

Challenge	Why it is hard	Active research
Qubit quality	Coherence/fidelity must be improved by another 10-100×	Materials science, new qubit types (fluxonium, bosonic)
Scaling	Wiring, power, cooling, crosstalk	Modular architectures, optical interconnects, cryo-CMOS
Error correction overhead	1000s of physical qubits per logical qubit	Better codes (qLDPC), better qubits
Connectivity	Nearest-neighbour limits circuits	Long-range couplers, shuttling ions, photonic links
Classical bottleneck	Decoding must keep up in real time (µs)	FPGA/ASIC decoders, AI decoders
Verification	How to check a result classical computers can't compute?	Cross-entropy benchmarking, interactive proofs
Workforce & cost	Interdisciplinary skills; expensive systems	Education, cloud access, national missions

★ *Remember*

Vision: million-qubit fault-tolerant computers solving chemistry and cryptography. Reality: few-hundred noisy qubits + first logical-qubit demos. Gap = error correction + engineering at scale.

9. Worked Examples

Example 1 - Gate budget

T_2 = 100 μs, gate time = 50 ns. Max operations before decoherence ≈ $100\times10^{-6} / 50\times10^{-9}$ = 2000 gates. Useful circuits needing 10^6 gates demand error correction.

Example 2 - Compound error

Per-gate fidelity 99.9 %, circuit with 500 gates: success ≈ 0.999^{500} ≈ 0.61 (61 %). With 99 % fidelity: 0.99^{500} ≈ 0.0066 (0.7 %). Shows why small improvements in fidelity matter greatly.

Example 3 - Overhead

If one logical qubit needs ~1000 physical qubits and an algorithm needs 4000 logical qubits, total ≈ 4 million physical qubits (+ ancilla/routing overhead).

10A. Benchmarking Progress: How Do We Measure a Quantum Computer?

Metric	Meaning
Number of physical qubits	Raw size - not enough by itself!
Gate fidelity	1 – error per gate; 2-qubit fidelity is the key number (≥ 99.9 % wanted)
Coherence times T_1, T_2	How long quantum information survives
Connectivity	Which qubits can directly interact
Quantum volume (IBM)	2^n where n = largest square random circuit (n qubits × depth n) run successfully
Algorithmic qubits (IonQ)	Number of usable qubits for algorithms
Randomized benchmarking	Estimates average gate error via long random gate sequences
Cross-entropy benchmarking	Compares sampling output with ideal simulation (Google)
Logical error rate	Error per logical operation after error correction - ultimate metric

★ *Lesson*

Ten high-quality qubits can outperform a hundred noisy ones. Always ask: how many qubits, how good, and how connected?

10B. Glossary for Unit 3

Term	Meaning
Transmon	Superconducting charge-insensitive qubit based on Josephson junction
Josephson junction	Two superconductors separated by thin insulator; gives nonlinear inductance
Fidelity	Closeness of actual state/gate to the ideal (1 = perfect)
Syndrome	Outcome of parity checks that signals which error occurred
Ancilla	Helper qubit used for measurement/error checks
Logical qubit	Encoded, error-protected qubit
Dilution refrigerator	Cryostat reaching ~10 mK using He-3/He-4 mixture
Crosstalk	Unwanted influence between neighbouring qubits/controls
Leakage	Qubit leaves the computational $\{\|0\rangle, \|1\rangle\}$ subspace
Fault tolerance	Computing reliably with unreliable components
Transpilation	Compiling a circuit to a device's native gates and connectivity

11. Short Answer Practice (Model Answers)

Q1. Why can't we just copy qubits for error correction?

No-cloning forbids copying an unknown state. Instead we spread the information of one qubit over an entangled state of many qubits ($\alpha\,|000\rangle + \beta\,|111\rangle$) and measure only parities (syndromes), which reveal errors but not α or β.

Q2. What is the difference between physical and logical qubits?

A physical qubit is the actual hardware two-level system (noisy). A logical qubit is an error-protected qubit encoded across many physical qubits, with much lower error rate. Useful algorithms are written in logical qubits.

Q3. Why do superconducting qubits need ~15 mK?

Thermal energy must be far below the qubit energy gap ($hf \approx 0.24$ K for 5 GHz) otherwise thermal photons randomly excite the qubit; also superconductivity of aluminium/niobium needs low temperature.

Q4. Compare quantum hardware in one line each

Superconducting - fast & fabricable; Ion - accurate & connected; Photonic - networkable & room-temperature; Neutral atoms - large arrays; Spin-silicon - small & CMOS-compatible; Topological - protected but unproven.

Q5. What is the threshold theorem and why is it important?

If physical error rate $p < p_{th}$, increasing code size reduces logical error exponentially. So arbitrarily long computations become possible with polynomial overhead - the theoretical justification for building large quantum computers.

Q6. What does 'quantum software' do that classical software does not?

It must respect reversibility and no-cloning, map circuits onto limited connectivity, schedule pulses, suppress noise, and run classical decoders in real time, while hiding hardware details from the programmer.

Q7. What is NISQ?

Noisy Intermediate-Scale Quantum: 50-1000+ imperfect qubits, no full error correction. Good for experiments, benchmarks and hybrid heuristics; not enough for Shor on large keys.

★ *Unit 3 summary in 5 lines*

1. Need scalable, initializable, coherent, controllable, measurable qubits. 2. Environment destroys quantum states → isolation + cold + vacuum. 3. Entanglement of many qubits is extra fragile. 4. QEC (surface code) with threshold makes large computers possible but costs ~1000× qubits. 5. Platforms and software are racing to close the gap between vision and reality.

12. Revision Questions

1. *Explain the DiVincenzo criteria.*
2. *Why are quantum systems fragile? Discuss decoherence, noise and control errors.*
3. *State the four conditions for a functional quantum system.*
4. *Why is it difficult to maintain entanglement?*

5. *Explain why quantum error correction is necessary and how it works without cloning.*
6. *Compare superconducting, trapped-ion and photonic platforms.*
7. *Distinguish between vision and reality in quantum computing today.*
8. *Describe the role of quantum software.*

UNIT 4

Quantum Communication and Computing - Theoretical Perspective

Unit 4 at a glance

Quantum vs classical information · basics of quantum communication · QKD (BB84, E91) · entanglement in communication · quantum internet · intro to quantum computing · quantum parallelism · classical vs quantum gates · decoherence & error correction · real-world importance & future

1. Quantum vs Classical Information (Recap)

Classical	Quantum
Bit: 0 or 1	Qubit: $\alpha \mid 0\rangle + \beta \mid 1\rangle$
Can be copied, amplified, read many times	Cannot be cloned; reading disturbs it
Security rests on hard maths (RSA)	Security rests on laws of physics
Channel = wire, fibre, radio	Channel carries photons, preserving superposition
Noise → bit errors	Noise → bit & phase errors, decoherence

2. Basics of Quantum Communication

Quantum communication transfers quantum states (usually of photons) between distant parties, using the principles of superposition, measurement disturbance, no-cloning and entanglement. It enables tasks impossible classically - secure keys, teleportation, networking of quantum computers.

» Components of a quantum communication link

- *Source: produces single photons or entangled photon pairs (e.g. spontaneous parametric down-conversion, quantum dots).*
- *Encoder: writes information into polarization, phase, or time-bin.*
- *Quantum channel: optical fibre (loss ~0.2 dB/km at 1550 nm) or free space / satellite link.*
- *Classical channel: ordinary authenticated channel for basis comparison, error correction.*
- *Detector: single-photon detectors (APD, superconducting nanowire).*
- *Quantum memory & repeaters: store and relay quantum states over long distances.*

★ *Why photons?*

Move at light speed, barely interact with the environment (low decoherence), travel through fibre/air/space, easy to encode and measure. Drawback: they get lost (absorbed/scattered) - and a lost photon cannot be re-amplified because of no-cloning.

» Channel loss problem

Fibre loss grows exponentially: after 100 km (0.2 dB/km) only 1 % of photons survive; after 500 km only 10^{-10}. Classical signals are amplified every ~80 km; quantum signals cannot be amplified (no-cloning). Solutions: quantum repeaters (entanglement swapping + quantum memory + purification) or satellites (loss mostly in the first few km of atmosphere).

3. Quantum Key Distribution (QKD)

Secure communication needs a shared secret key. Classical key-exchange (RSA, Diffie-Hellman) relies on the assumed difficulty of factoring - broken by Shor's algorithm on a large quantum computer. QKD lets two parties (Alice & Bob) establish a random secret key whose security is guaranteed by quantum physics, and any eavesdropper (Eve) is detected.

» 3.1 BB84 protocol (Bennett & Brassard, 1984)

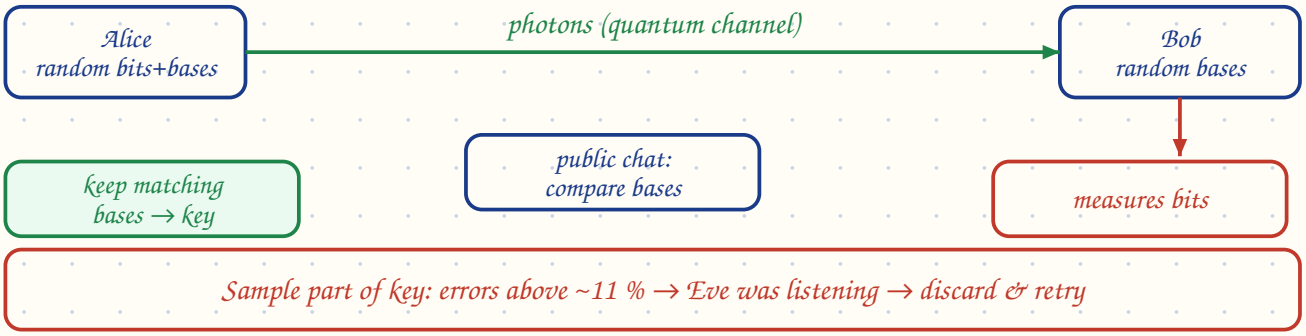


Fig 1: BB84 quantum key distribution

1. *Alice generates random bits and, for each bit, a random basis: rectilinear (+: H/V) or diagonal (×: D/A).*
2. *She sends single photons: bit 0/1 in + basis → H/V; bit 0/1 in × basis → D/A.*
3. *Bob measures each photon in a randomly chosen basis. If he picks the wrong basis, result is random.*
4. *Over a public channel they announce the bases (not the bits) and keep only positions where bases matched (~50 %). This is the sifted key.*
5. *They publicly compare a random sample. If the error rate (QBER) is low (< ~11 %), no significant eavesdropping; else abort.*
6. *Error correction + privacy amplification → final secret key; use it with one-time pad (perfect secrecy) or AES.*

Alice bit	1	0	1	1	0
Alice basis	+	×	×	+	×
Photon sent	V	D	A	V	D
Bob basis	+	+	×	×	×
Bob result	1	random	1	random	0
Keep?	✓	✗	✓	✗	✓

If Eve intercepts and measures in a random basis, she guesses wrong 50 % of the time; she then resends a wrong-state photon, which makes Bob's result wrong in 50 % of those cases → 25 % error in the sifted key. Eve is detected by the error rate. She cannot copy photons (no-cloning).

Other QKD protocols

B92: two non-orthogonal states. E91 (Ekert, 1991): uses entangled pairs; security from Bell-inequality violation. Decoy-state BB84: fixes weak-laser-pulse photon-number-splitting attacks (practical). CV-QKD: continuous variables, uses standard telecom components. MDI-QKD / Twin-field: remove detector vulnerabilities and extend distance.

» Practical issues

- Real sources are weak lasers, not perfect single-photon sources → decoy states.
- Detector side-channel attacks (blinding) → device-independent / MDI protocols.
- Distance limited to ~100-400 km in fibre without repeaters; key rate falls with distance.
- Needs authenticated classical channel (pre-shared short key).
- Standards bodies (ETSI, ITU, ISO) and NSA/NCSC caution: QKD complements, but does not replace, post-quantum cryptography.

Exam tip / Remember

Be able to (i) draw BB84 flow, (ii) work a table of bits/bases/sifted key, (iii) explain why eavesdropping causes 25 % errors, (iv) mention no-cloning and measurement disturbance.

4. Role of Entanglement in Communication

- E91 QKD: a source sends one photon of each entangled pair to Alice and Bob. Matching-basis results are perfectly correlated → key. Bell test on remaining data certifies no eavesdropper (device-independent security).
- Quantum teleportation: transmit an unknown qubit using shared entanglement + 2 classical bits (see Unit 2). Basis for quantum networks and gate teleportation in computers.

- *Entanglement swapping*: if A-B and C-D are entangled pairs, a Bell measurement on B and C entangles A and D even though they never met. This is the core of a quantum repeater.
- *Superdense coding*: 2 classical bits per qubit using entanglement.
- *Entanglement distribution*: satellites (Micius: 1200 km), fibre links (Delft 1.3 km loophole-free; Chinese/Euro networks).
- *Distributed quantum computing / sensing*: entangled links let separate processors act as one.

> **Quantum repeater idea**
> Divide long distance into short segments. Create entanglement in each segment → store in quantum memories → purify (distil better entanglement from noisy pairs) → connect segments by entanglement swapping. Loss then grows only polynomially, not exponentially.

5. The Idea of the Quantum Internet

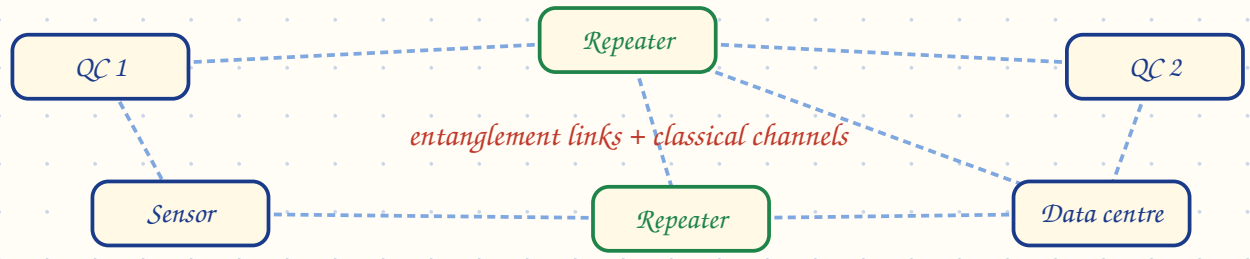


Fig 2: Sketch of a quantum internet

A *quantum internet* is a network that can distribute entanglement and transmit qubits between quantum devices anywhere in the world, running alongside the classical internet (which is still needed for control signals). Kimble (2008) laid out the vision; Wehner, Elkouss and Hanson (2018) described stages of its development.

Stage	Capability	Status
1. Trusted-repeater networks	QKD between nodes (nodes trusted)	Operational (Beijing-Shanghai, SECOQC, Tokyo QKD, India pilot links)
2. Prepare-and-measure networks	End-to-end QKD & secure comms without trusting nodes	Lab demonstrations
3. Entanglement distribution	Entangled pairs on demand between any two nodes	Early experiments (Delft, Hefei)
4. Quantum-memory networks	Storage, teleportation of qubits	Research
5. Few-qubit fault-tolerant networks	Distributed quantum computing, blind computing	Future

Stage	Capability	Status
6. Quantum computing networks	Full-scale network of QCs	Long-term

» Applications of the quantum internet

- Provably secure communication & key distribution at global scale.
- Secure access to cloud quantum computers (blind quantum computing - server cannot see your data or algorithm).
- Distributed quantum computing: link small processors to make a big one.
- Clock synchronization and networks of telescopes / sensors with improved precision.
- Quantum voting, secret sharing, secure multi-party computation, quantum money.

> Important projects
> India: ISRO free-space QKD demos (300 m → 1 km), DRDO-IIT Delhi 100 km QKD, NQM satellite & inter-city targets. China: Micius satellite, 2000 km backbone. EU: EuroQCI. USA: DOE quantum internet blueprint (2020), Chicago network. Netherlands: QuTech Delft - Hague link. Singapore, Japan, South Korea, UK networks.

6. Introduction to Quantum Computing

A quantum computer uses qubits and quantum gates to process information. The main idea (Feynman 1981, Deutsch 1985): since nature obeys quantum mechanics, a computer built from quantum components can simulate nature - and some mathematical problems - far more efficiently than classical computers.

- Input state is prepared in $|00\ldots0\rangle$.
- A sequence of quantum gates (unitary operations) creates superpositions, entanglement and interference.
- Finally, qubits are measured to obtain a classical bit string; algorithm is repeated many times to estimate the answer's probability.

> Famous algorithms
> Shor's algorithm (1994): factoring in polynomial time → breaks RSA/ECC. (Exponential speed-up.)
> Grover's algorithm (1996): search among N items in $\sim\sqrt{N}$ steps (quadratic speed-up).
> Quantum simulation / VQE / QPE: chemistry & materials.
> HHL: linear systems; QAOA: optimization; Quantum machine learning (under study).
> Deutsch-Jozsa, Simon: early proofs of quantum advantage.

7. Quantum Parallelism (Many States at Once)

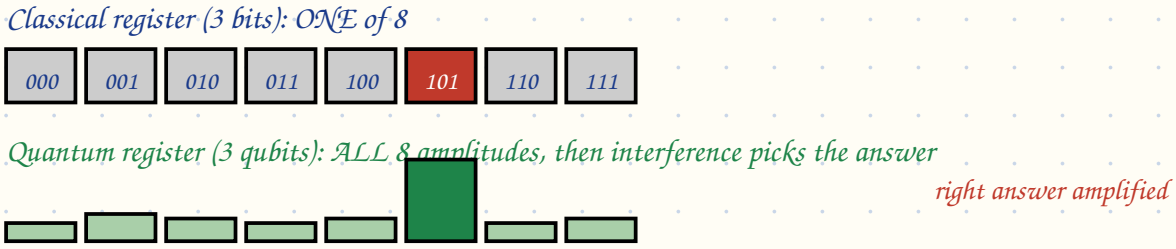


Fig 3: Superposition of all inputs - then interference

Apply Hadamard gates to n qubits ($|0\rangle^{\otimes n}$) → equal superposition of all 2^n inputs: $(1/\sqrt{2^n})$ $\Sigma\,|x\rangle$. Applying a function $f(x)$ implemented as a unitary U_f acts on every term simultaneously: $\Sigma\,|x\rangle\,|f(x)\rangle$. This is quantum parallelism.

★ *But there is a catch!*

Measurement returns only ONE random x and $f(x)$. The advantage comes from interference: algorithms are designed so wrong answers cancel and right ones add. Without clever interference there is no speed-up. Quantum parallelism ≠ 'brute force on all cases and read all answers'.

Deutsch's problem (smallest example)

Given $f:\{0,1\}\to\{0,1\}$, is f constant or balanced? Classically: 2 evaluations. Quantum: 1 evaluation (using superposition and interference) - the first demonstration that a quantum algorithm can beat classical.

Problem	Best classical	Quantum	Speed-up
Factoring N	sub-exponential (GNFS)	Shor: polynomial in log N	Exponential (superpolynomial)
Unstructured search (N items)	$O(N)$	Grover: $O(\sqrt{N})$	Quadratic
Simulating quantum systems	Exponential in particle number	Polynomial	Exponential
Sorting / basic arithmetic	Fast already	No advantage	None

8. Classical Gates vs Quantum Gates

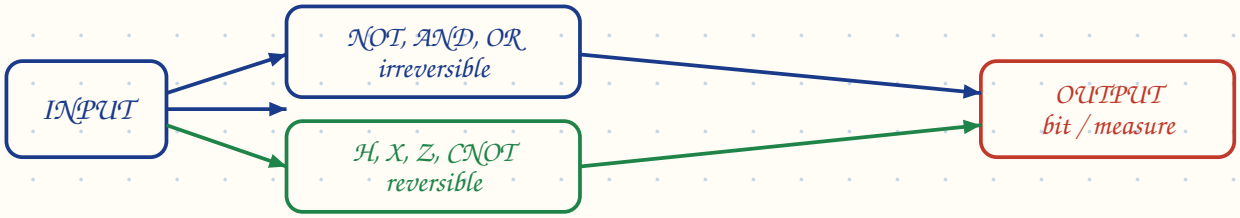


Fig 4: Irreversible classical logic vs reversible quantum gates

Classical gate	Quantum gate
NOT, AND, OR, NAND, XOR	X, Y, Z, H, S, T, CNOT, CZ, Toffoli
Irreversible (AND loses information, generates heat)	Reversible (unitary); no information lost
Acts on definite bit values	Acts on superpositions linearly
Truth table	Unitary matrix
NAND is universal	H + T + CNOT is universal (approximately)
Output always definite	Output is a state; measurement needed
Can copy bit (fan-out)	Cannot copy qubit (no-cloning)

» Important quantum gates

Gate	Effect	Matrix
X (NOT)	$\|0\rangle \leftrightarrow \|1\rangle$ (bit flip)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Z	$\|1\rangle \rightarrow -\|1\rangle$ (phase flip)	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Y	bit and phase flip	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
H (Hadamard)	$\|0\rangle \rightarrow (\|0\rangle + \|1\rangle)/\sqrt{2}$, $\|1\rangle \rightarrow (\|0\rangle - \|1\rangle)/\sqrt{2}$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
S, T	phase shifts of 90°, 45°	$\mathrm{diag}(1,i)$, $\mathrm{diag}(1,e^{i\pi/4})$
CNOT	Flip target if control = 1 (entangles)	4×4 matrix
Toffoli (CCNOT)	Flip target if both controls = 1	8×8; makes classical logic reversible

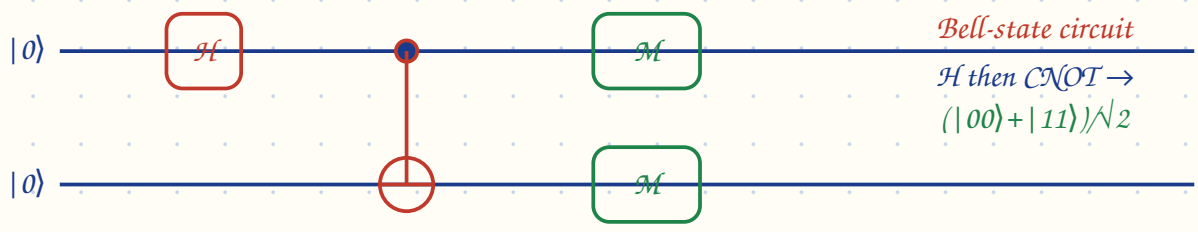


Fig 5: Circuit that creates a Bell state

Creating a Bell state

Start $|00\rangle \rightarrow H$ on qubit 1: $(|0\rangle + |1\rangle)|0\rangle/\sqrt{2} \rightarrow$ CNOT (control 1, target 2): $(|00\rangle + |11\rangle)/\sqrt{2}$. Entanglement made with just two gates!

9. Challenges: Decoherence and Error Correction

- *Qubits lose coherence in microseconds-seconds; circuits need thousands/millions of gates.*
- *Each gate has 0.1-1 % error; errors accumulate (see Unit 3).*
- *QEC encodes logical qubits in many physical qubits; surface code needs ~10^3 physical per logical.*
- *Fault-tolerant computing requires error rates below threshold, fast classical decoding, and fresh ancilla qubits.*
- *Quantum communication faces loss, noise and the lack of practical quantum memory/repeaters.*
- *Interfacing communication & computing: converting stationary qubits (matter) to flying qubits (photons) with high efficiency (transduction) is a research frontier.*

10. Real-World Importance and Future Potential

- *Cybersecurity: migrate to post-quantum cryptography (NIST standards: ML-KEM/Kyber, ML-DSA/Dilithium, SLH-DSA/SPHINCS+); combine with QKD for ultra-sensitive links (banking, defence, government).*
- *Drug & materials discovery: simulate molecules (e.g. FeMoco for fertilizer, battery electrolytes, catalysts).*
- *Optimization: logistics, scheduling, portfolio, energy grids.*
- *Secure cloud: blind quantum computing; quantum-safe cloud services.*
- *Science: tests of fundamental physics, distributed telescope arrays, quantum sensors networks.*
- *Future outlook: hybrid HPC + QPU data centres; 2030s early fault-tolerant machines; global quantum internet by 2040s (optimistic).*

Exam tip / Remember

Timeline hint: 1984 BB84 · 1991 E91 · 1994 Shor · 1996 Grover · 2016 Micius · 2019 'supremacy' · 2022 Nobel Prize (Aspect, Clauser, Zeilinger) · 2024 NIST PQC standards.

10A. Classical Cryptography vs Quantum Threats - Why It Matters

Cryptosystem	Basis	Threat from quantum computer
RSA, Diffie-Hellman, ECC	Factoring / discrete logarithm	Broken by Shor's algorithm
AES-128/256, SHA-2/3	Symmetric & hash	Grover halves effective key length → use AES-256
One-time pad	Random key as long as message	Information-theoretically secure; key distribution is the problem → QKD
Lattice, code, hash-based (PQC)	Hard problems believed quantum-resistant	Standardized by NIST (2024); software-only
QKD	Laws of physics	Secure against any computational power; needs hardware links

'Harvest now, decrypt later'
Adversaries can record encrypted traffic today and decrypt it once a quantum computer exists. Data that must stay secret for 20-30 years (health, defence, diplomatic) is already at risk - so migration to quantum-safe encryption must start now.

» Privacy amplification & one-time pad (QKD post-processing)

- After sifting, Alice and Bob run information reconciliation (e.g. Cascade/LDPC) to fix remaining errors.
- Then privacy amplification: hash the key to a shorter one so that whatever partial information Eve might have is reduced to almost zero.
- Final key used for one-time pad: ciphertext = message XOR key; unbreakable if key is random, used once and secret.

» Quantum random number generators (QRNG)

Classical computers produce pseudo-random numbers by algorithms. A QRNG measures a quantum event - e.g. which output a photon takes at a 50:50 beam splitter, or vacuum fluctuations - giving true randomness certified by physics. Commercial QRNG chips are already in smartphones (e.g. Samsung) and data centres; useful for cryptography, simulations and lotteries.

10B. Quantum Computing Models (Conceptual)

Model	Idea	Examples
Gate-based (circuit)	Sequence of unitary gates on qubits	IBM, Google, IonQ
Measurement-based (one-way)	Prepare big entangled cluster state; measurements drive the computation	Photonic (PsiQuantum, Xanadu)
Adiabatic / Annealing	Slowly evolve a system from simple to problem Hamiltonian; ground state = answer	D-Wave (optimization)
Topological	Information in non-local anyons → inherently error-resistant	Microsoft (research)
Analog quantum simulation	Engineer a quantum system that mimics another	Cold atoms, ion crystals

11. Solved Examples

Example 1 - Sifted key length

Alice sends 1000 photons, bases random. Bob's basis matches ~50 %: sifted key ≈ 500 bits. Sacrificing 100 bits for error check leaves ≈ 400 bits before privacy amplification.

Example 2 - Eavesdropper detection

Intercept-resend by Eve gives QBER ≈ 25 %. If Alice & Bob compare 100 bits, probability that Eve (attacking every photon) is undetected ≈ $(3/4)^{100} \approx 3\times10^{-13}$.

Example 3 - Grover speed-up

Search 1,000,000 items: classical ~ 500,000 checks on average; Grover ≈ $(\pi/4)\sqrt{10^6} \approx 785$ queries.

Example 4 - Fibre loss

200 km at 0.2 dB/km → 40 dB loss → fraction transmitted 10^{-4}. A 1 GHz photon source gives ~10^5 photons/s received.

12. Revision Questions

1. Compare quantum and classical information.
2. Describe the main elements of a quantum communication system.
3. Explain BB84 with an example table. How is eavesdropping detected?
4. Role of entanglement: E91, teleportation, entanglement swapping.

5. *What is a quantum repeater and why is it needed?*
6. *What is the quantum internet? Describe its stages and applications.*
7. *Explain quantum parallelism and its limitation.*
8. *Compare classical and quantum gates; construct a Bell-state circuit.*
9. *Why are decoherence and error correction challenges?*
10. *Discuss the real-world importance of quantum communication and computing.*

13. Short Answer Practice (Model Answers)

Q1. Why is BB84 secure?

Because (i) a single photon cannot be copied (no-cloning), (ii) measuring in the wrong basis randomizes and disturbs the state, and (iii) Alice and Bob can detect the resulting errors by comparing a sample of their key. Any eavesdropping is revealed as an abnormally high QBER.

Q2. QKD vs post-quantum cryptography (PQC)?

QKD: security from physics, needs special hardware & links, limited distance, key exchange only. PQC: software algorithms believed hard even for quantum computers, deployable anywhere, security based on unproven assumptions. Best practice: use both where needed.

Q3. Why can quantum signals not be amplified?

An amplifier would need to copy the photon's unknown state - forbidden by no-cloning. Hence repeaters use entanglement swapping and quantum memories rather than amplification.

Q4. What is quantum advantage vs supremacy?

Supremacy: solving any task a classical machine cannot in reasonable time. Advantage: doing a useful task better than classical methods. Supremacy demos (Google 2019, USTC 2020) are contrived; practical advantage is still pending.

Q5. Why is the Hadamard gate important?

It converts basis states to superpositions ($|0\rangle \rightarrow |+\rangle$) and vice-versa, enabling parallelism and interference. H on all qubits creates the uniform superposition used by Deutsch-Jozsa, Grover and Shor.

Q6. Entanglement swapping in 3 lines

Pairs A-B and C-D are entangled. A Bell measurement on B and C is performed and the result is told to the ends. Now A and D are entangled even though they never interacted - the building block of quantum repeaters.

★ *Unit 4 summary in 5 lines*

1. Quantum communication moves quantum states (photons). 2. QKD (BB84/E91) gives physics-based security. 3. Entanglement → teleportation, swapping, repeaters. 4. Quantum computing = qubits + unitary gates + interference; Shor and Grover are star algorithms. 5. Decoherence and loss are the big hurdles; quantum internet is the long-term goal.

UNIT 5

Applications, Use Cases and the Quantum Future

Unit 5 at a glance

Healthcare & drug discovery · materials science · logistics & optimization · quantum sensing & timing · industry case studies (IBM, Google, Microsoft, PsiQuantum) · ethics, society, policy · adoption challenges: cost, skills, standards · careers · education/research & India's opportunity

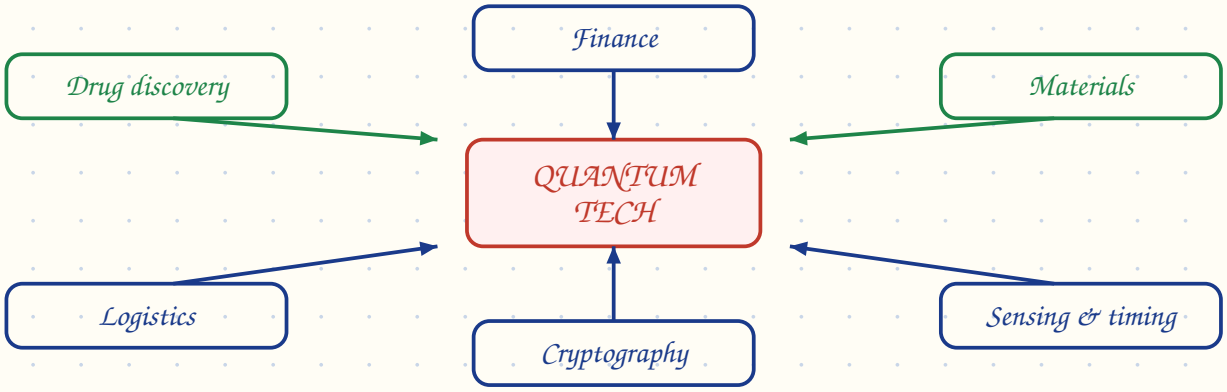


Fig 1: Where quantum technologies are expected to create value

1. Real-World Application Domains

Quantum technologies are not meant to replace classical computers. They target specific problems where quantum effects give an advantage: simulation of nature, optimization/search, secure communication and precision measurement. Most near-term uses are hybrid: classical computers + quantum accelerators (QPUs), like GPUs.

» 1.1 Healthcare and drug discovery

Drugs work through molecular interactions governed by quantum mechanics. Classical computers cannot exactly simulate molecules beyond ~tens of electrons because the state space grows exponentially. A quantum computer represents the molecule's electronic structure naturally.

- Molecular simulation: compute ground-state energies and reaction pathways with high accuracy (algorithms: VQE, Quantum Phase Estimation).
- Protein folding & binding: predicting how a drug candidate docks to a target; quantum-enhanced ML for screening huge chemical spaces.
- Faster, cheaper R&D: average drug costs > US$2 billion and ~10-15 years; better in-silico screening could cut failures.
- Personalized medicine & genomics: optimizing treatment plans, analysing genomic data, radiotherapy planning (optimization).

- *Medical imaging & diagnosis: quantum sensors - MEG with optically pumped magnetometers, NV-diamond MRI, improved MRI contrast.*
- *Collaborations: Roche, Boehringer Ingelheim with Google/Cambridge Quantum; Merck, Pfizer, AstraZeneca pilot projects; Moderna with IBM (mRNA).*

» 1.2 Material science

- *Design of new materials by simulation: high-temperature superconductors, better batteries (lithium-ion/solid-state electrolytes), solar-cell materials, light-weight alloys.*
- *Catalysis: understanding the nitrogenase enzyme (FeMoco) could enable energy-efficient fertilizer production (Haber-Bosch uses ~1-2 % of world energy); carbon capture catalysts.*
- *Condensed-matter models (Hubbard model, spin liquids) - analog quantum simulators with cold atoms already beat classical in some regimes.*
- *Quantum materials for quantum devices: superconductors, diamond, silicon-28, topological materials - a virtuous loop.*
- *Companies: BASF, Daimler/Mercedes (batteries), Boeing/Airbus (materials), Toyota.*

> ★ *Why quantum helps chemistry*
> *Electron correlation is intrinsically quantum. N spin-orbitals require 2^N amplitudes classically; N qubits store them directly. Rough estimate: ~100-200 logical qubits for useful small molecules, ~4000 logical qubits for FeMoco-scale problems.*

» 1.3 Logistics and optimization

- *Many business problems = choose the best among astronomically many options (travelling salesman, vehicle routing, scheduling, portfolio selection, supply-chain, flight gates, power-grid balancing).*
- *Quantum approaches: QAOA, quantum annealing (D-Wave), Grover-based search, hybrid solvers.*
- *Pilot cases: Volkswagen traffic-flow optimization (Lisbon), DHL/Airbus route & loading, Port of Los Angeles scheduling, Goldman Sachs/JPMorgan option pricing and risk (Monte-Carlo with quadratic speed-up), energy-grid optimization.*
- *Honest status: advantage not yet proven for real-sized problems; strong classical heuristics compete. Value may appear when error-corrected machines arrive.*

» 1.4 Quantum sensing and precision timing

Quantum sensing uses quantum states (superposition/entanglement) that are extremely sensitive to external fields, giving precision beyond classical limits (standard quantum limit → Heisenberg limit). It is the most mature quantum technology.

Sensor	Principle	Applications
Atomic clocks (optical lattice)	Frequency of atomic transition	GPS/GNSS, telecom sync, finance time-stamping, tests of relativity; error ~1 s in billions of years
NV-centre magnetometers	Spin of nitrogen-vacancy defect in diamond	Nanoscale MRI, brain & heart signals, battery inspection, mineral exploration
SQUIDs / OPMs	Superconducting loops / atomic vapours	Magnetoencephalography (MEG), geophysics
Atom interferometers	Matter-wave interference of cold atoms	Gravimeters & gradiometers: underground mapping, oil/water, inertial navigation without GPS
Quantum LiDAR / radar / imaging	Entangled or single photons	Stealth detection, imaging through fog, low-light bio-imaging
Quantum gravimeters	Cold-atom free-fall	Tunnel detection, volcano monitoring, resource exploration

Precision timing

The SI second is defined by caesium-133 (9,192,631,770 oscillations). Optical clocks (Sr, Yb, Al^+) are 100× more stable: they would neither gain nor lose a second in the age of the universe; they enable chronometric geodesy (measure height differences of cm by gravitational redshift).

2. Industrial Case Studies

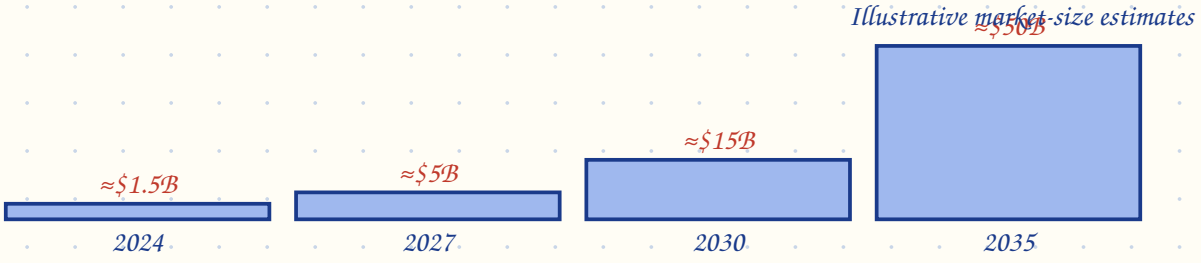


Fig 2: Rough growth of quantum market forecasts (illustrative)

Company	Approach / technology	Highlights
IBM	Superconducting transmon qubits; Qiskit; IBM Quantum Network; Quantum System Two (modular)	Eagle 127q (2021), Osprey 433q (2022), Condor 1121q (2023); Heron 133/156q higher-quality chips; roadmap to error-corrected 'Starling' (~200 logical qubits, 2029); qLDPC codes; cloud access to 100+ devices; Quantum-safe cryptography contributions; partnerships with IITs/IISc & Indian cloud hubs.

Company	Approach / technology	Highlights
Google Quantum AI	Superconducting; Cirq; Sycamore → Willow	2019 'quantum supremacy' (200 s vs 10,000 years claim); 2023 logical qubit experiments; Dec 2024 Willow 105q: exponential suppression of errors with larger surface code - 'below threshold'; random-circuit sampling in 5 minutes; 2025 verifiable 'Quantum Echoes' advantage claim; goal: useful error-corrected computer by ~2029-2030.
Microsoft	Topological qubits (Majorana); Azure Quantum cloud; Q# language	Partners with Quantinuum, Atom Computing for logical-qubit demos (2024); Majorana 1 chip announcement (2025, debated by experts); resource-estimator tool; quantum-safe crypto; Azure Quantum Elements (chemistry/AI).
PsiQuantum	Photonic, fusion-based, manufactured in GlobalFoundries silicon-photonics fab	Raised > US$1 billion (2025); building utility-scale million-qubit machines in Brisbane (Australia) & Chicago; pursuing cryogenic (~4 K) photonic modules.
Others	IonQ & Quantinuum (ions), Rigetti, D-Wave (annealing, claim of advantage on material simulation 2025), QuEra/Pasqal/Atom Computing (neutral atoms), Xanadu, Intel, Amazon (AWS Ocelot cat-qubit chip), NVIDIA (CUDA-Q hybrid)	India: QpiAI (25-qubit 'Indus' superconducting computer), TCS, L&T, Tata, Mahindra, BosonQ Psi, Pixxel/ISRO collaborations.

Exam tip / Remember

For exam: pick two companies, write 5 lines each - platform, key milestone (qubits/year), software (Qiskit/Cirq/Q#), roadmap. Also state that milestone claims are debated and evolving.

3. Ethical, Societal and Policy Considerations

» Ethical & societal issues

- Security & privacy: Shor's algorithm threatens present encryption; 'harvest now, decrypt later' surveillance risk; need quantum-safe migration.
- Digital divide / quantum divide: only wealthy nations and companies may access quantum power → inequality; open cloud access and education help.
- Dual use: military applications (code-breaking, navigation, stealth detection, sensing of submarines) → arms-race and export controls.

- *Economic disruption & jobs: new industries but possible disruption in finance, cybersecurity, pharma.*
- *Responsible innovation: anticipate misuse early (lesson from AI); inclusive governance; transparency vs hype ('quantum-washing', exaggerated claims).*
- *Environmental aspects: cryogenics use energy and helium-3; but quantum may reduce the energy needed for some computations and chemistry (fertilizer, batteries).*
- *Ethics of access & ownership: patents, open-source standards, data sovereignty.*

» Policy landscape

Policy area	Examples
National strategies	India NQM (2023), US NQI Act (2018) & CHIPS/Quantum acts, EU Quantum Flagship & Strategy (2023), UK 10-year plan, China 14th Five-Year Plan, Japan, Korea, Australia, Canada
Export controls	US/EU/UK controls on advanced quantum computers & components (2024), Wassenaar-style arrangements
Quantum-safe migration	NIST PQC standards (FIPS 203/204/205, Aug 2024); US NSM-10 / 2035 deadline; India's NCIIPC/DST advisories; CNSA 2.0
Standards	ISO/IEC JTC 3, IEEE P7130, ETSI QKD specs, ITU-T; BIS India working groups
Workforce	Fellowships, quantum education acts, industry-academia hubs
Ethics frameworks	UNESCO/OECD/WEF 'Quantum for Society' & responsible quantum technology principles

4. Challenges to Adoption

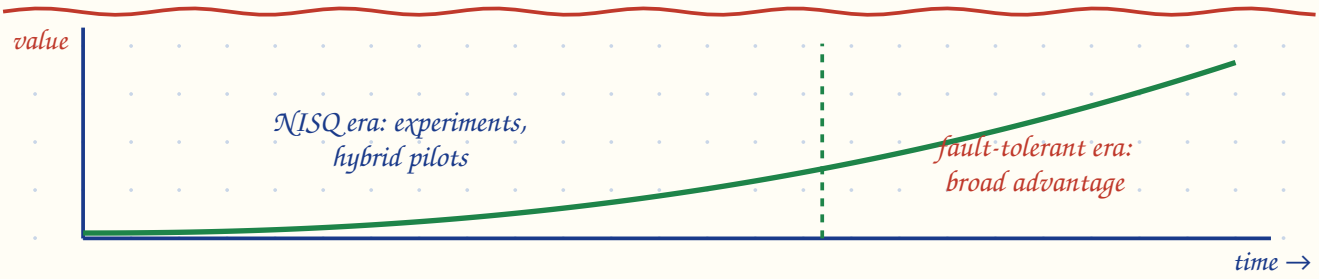


Fig 3: Value expected to rise sharply only after fault tolerance

Challenge	Details	Way forward
Cost	A superconducting system costs US$10-60 million; dilution fridges, lasers, cleanrooms; running costs and helium	Cloud access (pay-per-use), shared national facilities, modular/room-temperature designs, economies of scale

Challenge	Details	Way forward
Skills / talent	Interdisciplinary: physics + CS + engineering; global shortage (≈3 jobs per qualified candidate)	New degrees, NPTEL/NQM courses, bootcamps, upskilling programmes, industry internships
Standardization	No common benchmarks, gate sets, APIs, security certification for QKD/PQC	ISO/IEC, IEEE, BIS; open-source frameworks; benchmarks (quantum volume, CLOPS, algorithmic qubits)
Technical immaturity	Noise, small scale; unclear 'killer app'	Hybrid algorithms, error mitigation, co-design
Integration	Linking QPUs with HPC & cloud; software stacks	Quantum-HPC centres, middleware, CUDA-Q, Qiskit Runtime
Hype & trust	Unrealistic promises	Transparent roadmaps, independent verification
Supply chain	Helium-3, rare isotopes, specialized lasers, superconducting cables	Domestic manufacturing, material recycling

5. Emerging Careers in Quantum

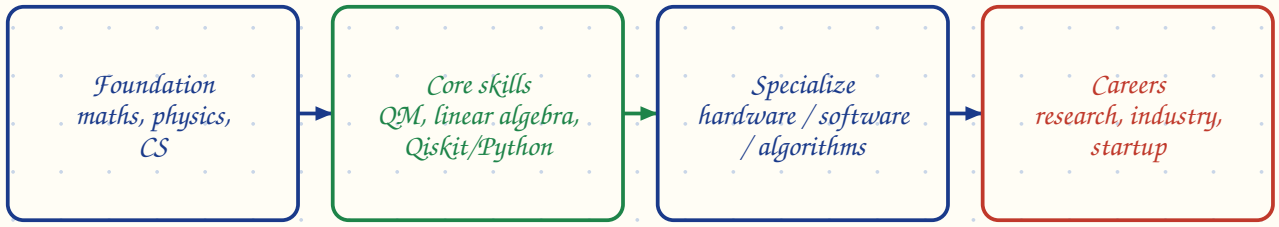


Fig 4: A pathway into quantum technology

Role	What they do	Skills
Quantum research scientist / physicist	Develop theory, qubits, protocols	QM, many-body physics, maths, papers; PhD usual
Quantum hardware / device engineer	Fabricate & test qubits, cryogenics, lasers, photonics chips	Microfabrication, RF/microwave, optics, vacuum, materials
Control & electronics engineer	FPGA, pulse generation, cryo-CMOS, readout chains	Digital/analog design, signal processing
Quantum software developer	Compilers, SDKs, cloud platforms, simulators	Python, C++/Rust, Qiskit/Cirq/PennyLane, software engineering
Quantum algorithm / application scientist	Map industry problems to quantum algorithms; resource estimation	Linear algebra, complexity, domain knowledge (chem, finance)

Role	What they do	Skills
Quantum cybersecurity specialist	PQC migration, QKD deployment, crypto-agility	Cryptography, networks, standards
Quantum network / communication engineer	Photonic links, repeaters, satellites	Optical comms, photonics, protocols
Quantum sensing engineer	Atomic clocks, magnetometers, navigation	Atomic physics, metrology, instrumentation
Product / business / policy roles	Strategy, partnerships, standards, ethics	Tech literacy + business/legal; communication

» Preparation pathway

1. *Foundations:* calculus, linear algebra, probability, classical & modern physics, programming (Python).
2. *Core quantum:* quantum mechanics, quantum information (Nielsen & Chuang; Preskill's notes; Qiskit Textbook; edX/NPTEL/Coursera courses).
3. *Hands-on:* run circuits on IBM Quantum / Amazon Braket / Azure Quantum; Qiskit, Cirq, PennyLane; open-source contributions; hackathons (Qiskit Fall Fest, IBM Quantum Challenge, NQM hackathons).
4. *Specialize:* hardware (cryogenics, photonics), algorithms (chemistry, optimization), software, communication.
5. *Higher studies / experience:* MSc/MTech/PhD at IISc, IITs, TIFR, IISERs, RRI; internships at QpiAI, TCS, IBM, Quantinuum, national labs.
6. *Soft skills:* interdisciplinary teamwork, communication, ethical awareness; certifications (e.g. IBM Quantum Developer Certification).

> *Exam tip / Remember*
> Quantum workforce needs are not only PhDs: technicians, software engineers, cyber specialists and project managers are required. A good mix: strong CS/maths + basic quantum knowledge = employable fast.

6. Educational and Research Landscape; India's Opportunity

» Global research landscape

- Leaders in publications/patents: USA, China, Europe (Germany, Netherlands, UK, France), Japan, Canada, Australia; hubs: MIT, Harvard, Caltech, Delft, Oxford, ETH Zurich, USTC, Waterloo's IQC.

- Public + private investment: > US$40 billion pledged worldwide; venture funding > US$2 billion/yr; 400+ startups.
- Education: MSc/PhD programmes, EU Quantum Technology Master's, Q-12 education initiatives in USA, open online courses, summer schools.

» India's landscape

Institution / initiative	Role
National Quantum Mission (DST, 2023-31, ₹6003 cr)	Four T-Hubs; build 50-1000 qubit computers; 2000 km QKD; sensors, materials; goal: make India a leading nation in QT by 2031
IISc Bengaluru	Quantum computing hub; superconducting & photonic research; Quantum Technology Initiative
IIT Madras (C-QuICT)	Quantum communication hub; QKD, post-quantum crypto
IIT Bombay	Quantum sensing & metrology hub
IIT Delhi	Quantum materials & devices hub
TIFR Mumbai	Superconducting qubits - India's first indigenous 6-qubit/ transmon work; fundamental research
RRI Bengaluru, IISER Pune/Mohali, IIT Kanpur, IIT Kharagpur, IIT Hyderabad, C-DAC	Quantum optics, communication, algorithms, supercomputing & quantum simulators
ISRO, DRDO, DAE	Satellite quantum communication; QKD links; defence applications
Amaravati Quantum Valley (Andhra Pradesh)	State initiative with IBM Quantum System Two; planned to host a 156-qubit machine & ecosystem; startups & skilling
Startups & industry	QpiAI (Bengaluru), BosonQ Psi, QNu Labs (QKD), Quantum Hub; TCS, Infosys, Wipro, L&T, Tata, Mahindra

» India's opportunity

- Talent advantage: one of the world's largest STEM and IT workforces; strong maths/physics tradition (Bose - Bose-Einstein statistics, Raman effect, Sen, Bhabha).
- Software & services strength: Indian IT firms can lead in quantum software, algorithms, cloud and services.
- Cost advantage & scale: lower-cost manufacturing and test infrastructure; large domestic market for secure communication (banking, defence, space).
- Policy push: NQM, ISRO satellite capability, semiconductor mission synergy, startup ecosystem (Startup India).
- Strategic autonomy: indigenous QKD/PQC and sensing reduce dependence on foreign tech.

- *Gaps: fabs & cryogenic supply chain, specialised labs, funding scale (small relative to USA/China), faster industry-academia linkage, retention of talent ('brain-drain').*

> *The big picture*
> *Quantum is a once-in-a-generation platform shift (like semiconductors and AI). Nations that invest early in people, standards and supply chains will shape the rules. India has the talent and momentum - execution and speed are the keys.*

6A. More Application Areas

» Finance

- *Monte-Carlo & option pricing: quantum amplitude estimation gives a quadratic speed-up (error ~1/N instead of 1/√N).*
- *Portfolio optimization, credit scoring, fraud detection, risk analysis (JPMorgan, Goldman Sachs, HSBC, Wells Fargo, Indian banks exploring).*
- *Quantum-safe banking: QKD links between data centres; PQC for UPI/SWIFT-like systems.*

» Energy, climate and environment

- *Better batteries, solar absorbers, fusion-plasma simulation, carbon-capture sorbents.*
- *Smart-grid optimization and renewable forecasting; quantum gravimeters to find water/minerals; climate-model acceleration (long-term).*
- *Fertilizer: cheaper ammonia through catalyst design (FeMoco) → less CO_2.*

» Artificial Intelligence & machine learning

- *Quantum machine learning: quantum kernels, variational classifiers, generative models - speculative, benefits not yet established.*
- *AI helps quantum: decoders for error correction, calibration, circuit compilation, discovering new algorithms.*
- *Hybrid pipelines: GPUs + QPUs (NVIDIA DGX Quantum, CUDA-Q).*

» Aerospace, defence & navigation

- *GPS-free inertial navigation using cold-atom accelerometers - for submarines, aircraft, autonomous vehicles.*
- *Secure satellite links, quantum radar/LiDAR, detection of hidden objects by gravity/magnetic anomalies.*
- *Concerns: dual-use & escalation; international norms needed.*

Domain	Technology	Timeframe (approx.)
Timing, magnetometry, gravimetry	Quantum sensing	Now - 5 years
QKD, QRNG, PQC migration	Quantum communication / security	Now - 10 years
Chemistry & materials small-scale	Quantum computing (NISQ + early FT)	5 - 10 years
Optimization, finance, large simulations	Fault-tolerant computing	10 - 20 years
Quantum internet, distributed QC	Networks + memory + repeaters	15 - 25 years

6B. Glossary for Unit 5

Term	Meaning
Quantum advantage	A quantum device performs a useful task better (faster/cheaper/more accurately) than any classical method
Quantum supremacy	Quantum device solves some (possibly useless) problem infeasible for classical computers
NISQ	Noisy Intermediate-Scale Quantum devices (50-1000 imperfect qubits)
Logical qubit	Error-protected qubit made from many physical qubits
Quantum volume	IBM's single-number benchmark of overall quality & size of a device
Hybrid algorithm	Algorithm splitting work between classical and quantum processors (VQE, QAOA)
PQC	Post-quantum cryptography: classical algorithms resistant to quantum attack
Crypto-agility	Ability to swap cryptographic algorithms quickly
Dual-use	Technology with civilian and military uses
T-Hub	Thematic Hub of the Indian National Quantum Mission
QaaS	Quantum computing as a Service via cloud

6C. Quick Timeline of Quantum Milestones

Year	Milestone
1900-1927	Planck, Einstein, Bohr, de Broglie, Schrödinger, Heisenberg - quantum mechanics born
1947-60	Transistor and laser - first quantum revolution
1981-85	Feynman & Deutsch propose quantum computers
1984, 1991	BB84 and E91 quantum cryptography
1994-96	Shor's and Grover's algorithms; Shor's error-correcting code (1995)
1998-2007	First few-qubit NMR, ion-trap and superconducting qubits
2016-19	Micius satellite; IBM Q Experience; Google 'supremacy' (2019)
2022	Nobel Prize in Physics - Aspect, Clauser, Zeilinger (entanglement experiments)
2023	India's National Quantum Mission; IBM Condor 1121 qubits
2024	NIST PQC standards; Google Willow below-threshold error correction
2025	UN International Year of Quantum Science & Technology; new logical-qubit and cat-qubit chips

7. Case-Study Style Mini Answers

Q. How can quantum computing speed up drug discovery?

(1) Simulate electronic structure of target molecules accurately (VQE, QPE) → better binding-energy estimates. (2) Reduce wet-lab trials by in-silico screening. (3) Hybrid quantum-classical ML for candidate selection. Present status: small molecules (H_2, LiH, BeH_2) done on hardware; practical benefit expected with error-corrected machines.

Q. Why are quantum sensors the nearest-term technology?

They need only a few well-controlled quantum systems (not millions of qubits), operate on short timescales, and gain from sensitivity not error-correction. Already commercial: atomic clocks, magnetometers, gravimeters.

Q. Discuss the challenges to adoption.

Cost (hardware, cryogenics), skills shortage, lack of standards, technical immaturity, integration with HPC, hype, supply chain. Mitigation: cloud access, education, standards bodies, hybrid algorithms, public funding, international collaboration.

Q. Is quantum computing a threat to current security?
Yes in the future: Shor can break RSA/ECC. Remedy: post-quantum cryptography (NIST 2024), crypto-agility, QKD for specific links; start migration now due to 'harvest now, decrypt later'.

8. Quick Revision & Likely Questions

1. *List the application domains of quantum technologies with examples.*
2. *How can quantum computers help in drug discovery and material science?*
3. *Explain how quantum sensing & atomic clocks work and their uses.*
4. *Discuss industrial efforts of IBM, Google, Microsoft and PsiQuantum.*
5. *Explain ethical, societal and policy issues related to quantum technology.*
6. *What are the barriers to adoption: cost, skills, standardization?*
7. *Describe career roles in quantum technology and the preparation pathway.*
8. *Explain India's opportunity and role in the global quantum race.*

★ *Final takeaway*
Quantum technologies are moving from labs to markets: sensing today, communication soon, computing next. Success depends on technology, talent, trust and policy together.